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Chapter 1

Introduction

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness is also responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

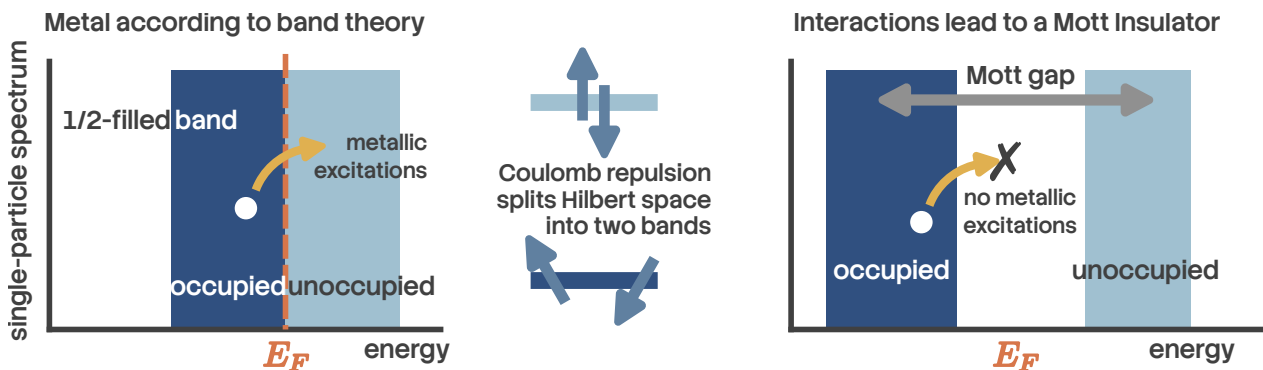


Figure 1.1

1.1 Metal-insulator transition arising from strong correlation

Initial attempts at explaining metal-insulator transitions was based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U ; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to a long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high- T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N. \quad (1.1)$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model

incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve d -electrons that are five-fold degenerate, and these orbitals can hybridise with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard. Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$). We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probability t . As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t , the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

Holon-doublon unbinding picture of the Mott MIT. In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in

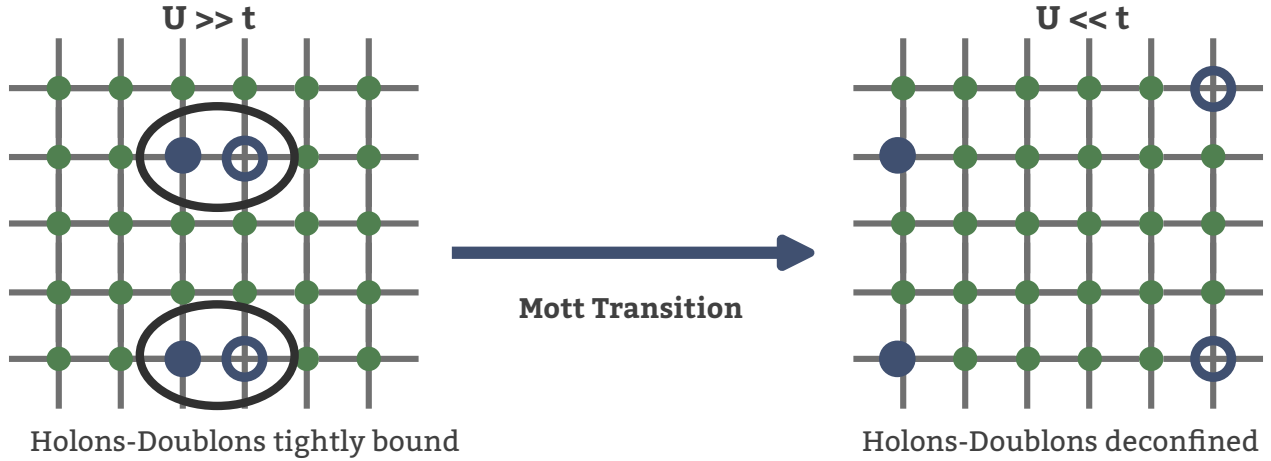


Figure 1.2

terms of the motion of holes and doubles in the background of a half-filled lattice (see Fig. 1.2). For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator (left, Fig. 1.2).

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double (right, Fig. 1.2). Current can then be passed by having the hole move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-filled honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust d -wave superconductivity and short-range magnetic order.

Spectral features of the Mott transition. One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\mathbf{k}, \sigma; t) = -i\theta(t)\langle\{c_{\mathbf{k},\sigma}(t), c_{\mathbf{k},\sigma}^\dagger\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem,

$$\frac{N}{V} = 2 \int_{G(0,\mathbf{k})>0} \frac{d\mathbf{k}}{(2\pi)^3}, \quad (2.1)$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of k -space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\mathbf{k}; \omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\mathbf{k}; \omega)$. The points in k -space where $\Sigma(\mathbf{k}; 0)$ vanishes is called a Luttinger surface. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [24]

$$G_R(i; \omega) = \frac{1+x}{\omega + \mu + U/2} + \frac{1-x}{\omega + \mu - U/2}, \quad (2.2)$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma} \rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1+x$ number of states while the upper band has $1-x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

1.3 Preliminary Insights into Mottness From a Toy Model

The Baskaran-Hatsugai-Kohmoto model [25, 26] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a Mott insulator. While its solvability offers an instructive method for studying Mott transition, the

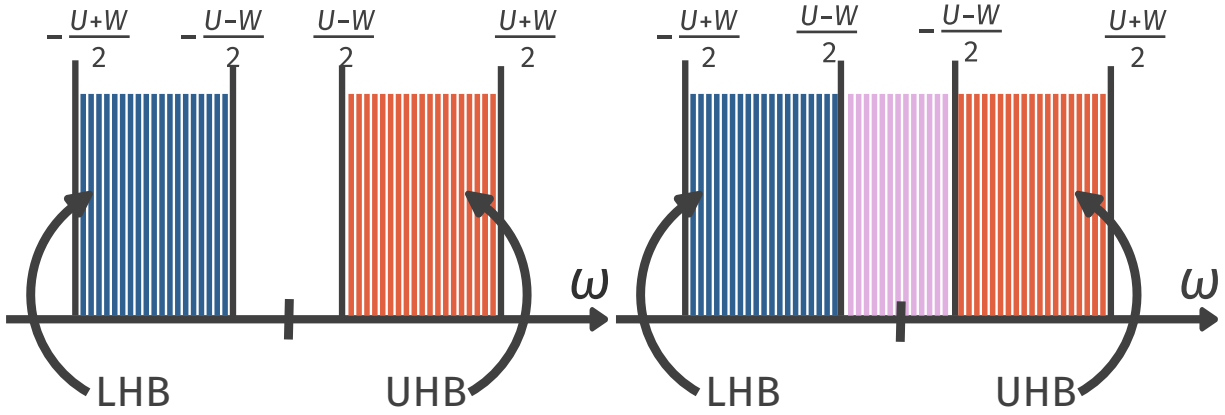


Figure 1.3

infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k}, \sigma} \epsilon(\mathbf{k}) n_{\mathbf{k}, \sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (3.1)$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k}, \sigma} - \frac{U}{2} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (3.2)$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: ($\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$, the other for adding an additional electron on a singly-occupied state: $\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$).

Upon combining states from other k -modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per k -state). Upon lowering the correlation strength U , the bands touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). For lower values of U , the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.4 Strong local correlations: models and materials

1.4.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [27]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (4.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum'_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (4.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

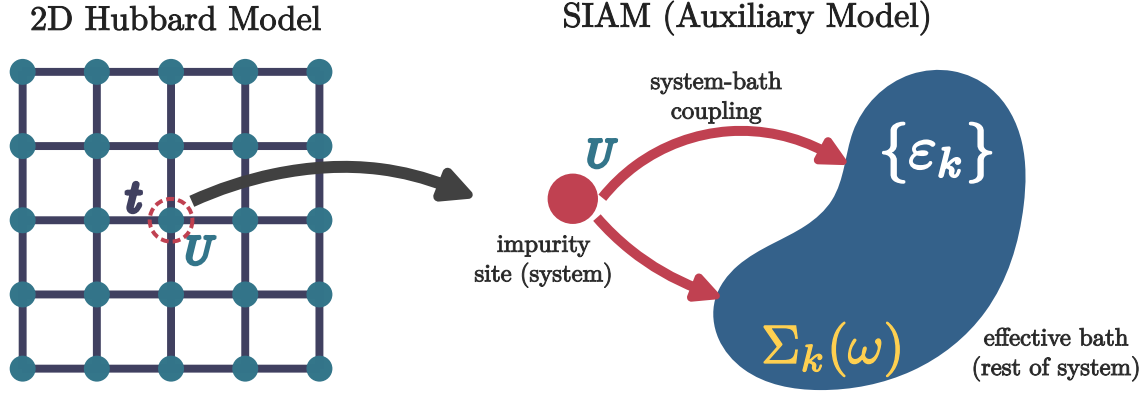


Figure 1.4: Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

1.4.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 28]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal–insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 29]

In order to motivate the impurity model–lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = - \sum_{i \neq j} J_{ij} S_i S_j - h \sum_i S_i = - \sum_i S_i \left(h + \sum_{j \neq i} J_{ij} S_j \right), \quad (4.4)$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j \rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H_{\text{Ising}}^{(\text{MF})} = - \sum_i (h + h_{\text{MF}}(i)) S_i, \quad (4.5)$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2\langle S_j \rangle J_{ij} . \quad (4.6)$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j \rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j \rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta \sum_{j \neq i} 2m_j J_{ij} + \beta h) . \quad (4.7)$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j . We therefore need an additional equation; this is provided by the so-called *self-consistency condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j . This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j \neq i} 2J_{ij} + \beta h) . \quad (4.8)$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 4.7 and 4.6, we have

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2m_j J_{ij} = \sum_{j \neq i} 2 \tanh(\beta(h_{\text{MF}}(j) + h)) J_{ij} . \quad (4.9)$$

The equation in terms of $h_{\text{MF}}(i)$ and $h_{\text{MF}}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

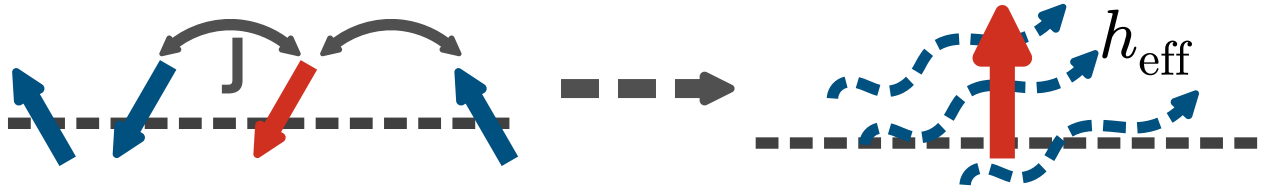


Figure 1.5

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the

formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 28].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = - \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4.10)$$

defined on a lattice with coordination number z .

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an “order parameter” and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k -space Green’s function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \quad (4.11)$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green’s function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \quad (4.12)$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes k -independent [28]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij} \Sigma_{\text{loc}}(\omega). \quad (4.13)$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \quad (4.14)$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = - \int_0^\beta d\tau d\tau' c^\dagger(\tau) G_{\text{MF}}^{-1}(\tau - \tau') c(\tau') + U \int_0^\beta d\tau n_\uparrow(\tau) n_\downarrow(\tau). \quad (4.15)$$

This acts as an auxiliary problem for the more complete problem locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green’s function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \quad (4.16)$$

The first term in eq. 4.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a “field” because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc} . As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc} , owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned} G_{\text{MF}} &= G_{\text{loc}}, \\ \Sigma_{\text{imp}}(\omega) &= G_{\text{MF}}^{-1}(\omega) - G_{\text{imp}}^{-1}(\omega), \\ \Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\ G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \end{aligned} \quad (4.17)$$

One can map various features of this self-consistent program with the set of equations obtained above for the Ising model:

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j .
- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.
- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.
- The fourth equation is equivalent, in the previous exercise, to using the computed “magnetisation” of site j in the second step to further compute the mean-field at site i .

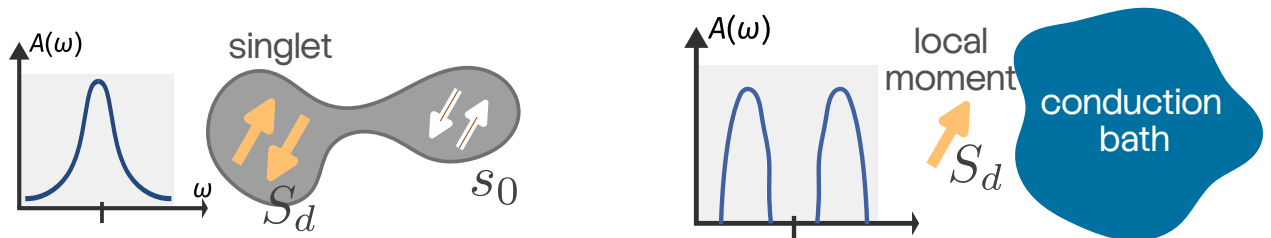


Figure 1.6

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. Importantly, this demonstrates the importance of quantum impurity models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.4.3 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d$ – or $4f$ –orbitals in such elements. This is because these orbitals are more more localised around the nucleus than the s – or p –orbitals of comparable energy, and hence lead to a reduced itinerant energy scale and amplify the effects of local correlations [29].

The localised nature of these orbitals can be understood as follows. For the $3s$ –orbital, the $1s$ and $2s$ orbitals have same orbital angular momentum l but lower principal quantum number n ; since all such wavefunctions must be orthogonal to each other, the $3s$ –orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d$ –orbital, there is no orbital with same l but lower n ; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f$ –orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d -shells in transition metals (e.g., Cu^{2+} , Mn^{3+} , V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U ; due to the small overlap between the localised d –orbitals, the d –electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [30], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d – d Coulomb interaction ($E_g \approx U$), or charge-transfer insulators, where the gap is dictated by the energy difference between the oxygen p -band and the metal d -band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy} , d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the

Fermi level lies within e_g . The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the d -orbitals and brings them closer to the p -orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a d -state to a p -state, compared to the d -level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the p -orbitals) also enhance the hybridisation between d - and p -orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ . In the lighter TMOs, the Fermi level passes through t_{2g} , and because they point away from the p -orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal-insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [31, 32]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [33–35]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [36–38]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [39–41]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^{2+} sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nickelates is different due to the strong coupling between the layers [42, 43]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [44].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2 , where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [45, 46]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal-insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [47, 48]. In prototypical compounds such as 1T-TaS₂, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [49–51]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal-insulator transition of 1T-TaS₂, demonstrating the macroscopic controllability of the Mott phase [52].

ARPES measurements in 2H-NbSe₂ and 2H-TaSe₂ partial gapping of the Fermi surface re-

stricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [53, 54]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [55]. Transport measurements on single crystals of TaSe₂ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [56]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine ‘blocking layers’ [57–60]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [61–63]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior (T –linear resistivity) to a Fermi liquid behaviour (T^2 –resistivity) [64]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [65].

ARPES measurements reveal a k –dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [66, 67]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [68, 69]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [70, 71].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [72, 73]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [74]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [75].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [76]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior (T –linear resistivity) and a vanishing Hall response, along with a di-

vergent effective quasiparticle mass as the transition is approached [77, 78]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [79, 80]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [81]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [82].

1.4.4 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch’s theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c_{k,\sigma}^\dagger d_\sigma + \text{h.c.}) + \epsilon_d \sum_{\sigma} n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} . \quad (4.18)$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U , the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [83]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [84]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1, k_2} \sum_{\alpha, \beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha, \beta} c_{k_1, \alpha}^\dagger c_{k_2, \beta} , \quad (4.19)$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: it makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c_{d\uparrow}^\dagger c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c_{d\downarrow}^\dagger c_{k_1\downarrow}$. This shows that the conduction electrons *are*

aware of the other electrons, and all of them must be treated together. Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson’s numerical renormalisation group approach [85, 86], density-matrix renormalisation group approach [87] or the method of *Bethe ansatz* [88, 89]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [90, 91]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K , the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [85].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [74, 86]. Away from half-filling, two other fixed points become available [74, 86]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [92–95]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations. At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U . The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [96], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger’s theorem [97, 98] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [99, 100].

Breakdown of Kondo screening While the vanilla Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to tweak it in order to introduce

Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [101]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c_{k_1,\alpha,\lambda}^\dagger c_{k_2,\beta,\lambda}, \quad (4.20)$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [101, 102]. Powerful methods like boundary conformal field theory [103–108], bosonization [109–113], numerical renormalization group [105, 114, 115] and Bethe ansatz [116–120] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [112, 121]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [101, 105, 111, 119, 122].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a parameter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as d -wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [123–126]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [124, 127–130]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [124, 131]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [126].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [132, 133]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [132–134] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [135]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge

excitations [136]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase with "topological order" to a screened Fermi liquid phase [137]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [138].

1.5 Phenomenology of Mott transition in Hubbard-like models

1.5.1 The Infinite-Dimensional Hubbard Model

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal-insulator transition (MIT) [4, 139–141]. While exact solutions exist in the limits of $d = 1$ [142] and $d = \infty$ [143], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle k -space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal-insulator transition at both zero and finite temperatures for this limiting case [18, 143–147]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2} , signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 145]. The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [145]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [148]. The quantum critical point itself (at $T = 0$) is likely to be an exotic state, and important questions about it remain unanswered. *Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?*

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the vanilla Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: *what are the minimal changes that one must implement into the Anderson impurity model in order to capture such a metal-*

insulator transition (1/2-filled Hubbard model on the Bethe lattice in $d = \infty$), in sufficient accuracy and precision? Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [146, 149]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [150]. These lines meet at a finite-temperature critical endpoint (U_c, T_c) , above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [146, 148, 151–153]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: *since a quantum impurity lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates?* Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [154, 155]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [154]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a “hidden” quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [156]. Luttinger’s theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [157]. The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: *What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?*

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U . In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a charge-transfer energy Δ for hopping from a d -orbital to a p -orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [158]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [159].

1.5.2 The two-dimensional Hubbard model

The 2D Hubbard model is one of the most well-studied models in condensed matter physics, particularly because of its paradigmatic role in Mott metal-insulator transition and its connections to the physics of copper oxide materials and d -wave superconductivity. Photoemission experiments reveal the presence of p and d bands close to the Fermi surface in such materials [39]; further work by Zhang and Rice showed that a specific combination of these orbitals governs the low-energy physics, leading to an effective description in terms of a single band Hubbard model [160]. The discovery of high-temperature superconductivity in nickelates and organic charge transfer salts has provided more platforms where the Hubbard model and its extensions can prove to be useful [161, 162].

While simple in appearance, the model has proved to be notoriously hard to solve except in highly specific parameter regimes, despite several decades of computational effort. This is due to several reasons [141]: (i) the presence of closely-lying energy scales arising from competing tendencies, (ii) the inability of several methods to either reach zero temperature or extrapolate to the thermodynamic limit, (iii) the tendency of approximate/finite-size methods to overestimate proclivities towards spontaneous symmetry breaking, and (iv) the presence of differing phases separated by slow crossovers without any sharp boundaries.

At half-filling, the 2D Hubbard model has a spectral gap to charge excitations for any non-zero interaction strength [163–167]. The behaviour below the charge gap is described by a nonlinear sigma model with parameters that depend on the interaction strength U [168]. Upon tuning to finite temperatures, the system goes through a sequence of crossovers from the insulating ordered state to a Fermi liquid state with coherent quasiparticles into a non-Fermi liquid state with partial gapping of the Fermi surface [165, 167, 169]. *Capturing the momentum dependence of static and dynamical quantities in sufficient detail remains an open challenge in such problems, since results from cluster methods suffer from sensitivity to periodisation schemes [170].*

Away from half-filling but at weak coupling, the Neel order is accompanied by incommensurate magnetic orders (also called spiral magnetic state) with wave vectors away from (π, π) , as seen from mean-field studies [171, 172]. Functional RG studies indicate that such solutions can exist up to fairly high doping if the superconducting instability is suppressed [173, 174]. Superconductivity with d -wave pairing has been achieved from renormalised perturbation studies as well as from diagrammatic Monte Carlo (diagMC) and functional RG (fRG) calculations [175–177]. More intricate competition between the superconducting and stripe states is found in three-band Hubbard models from variational Monte Carlo, cluster DMFT and DMRG calculations [178–180]. *Further work is necessary to resolve the closely-lying phases in such models.*

The phenomena of pseudogap (PG), where the Fermi surface acquires a momentum-dependent suppression of spectral weight, has also been observed above a certain temperature from methods like diagMC and fRG [167, 181]. The gap is seen to open first near the antinodal point $(\pi, 0)$ and spread across the Fermi surface until it reaches the node $(\pi/2, \pi/2)$. The mechanism of the pseudogap at weak to intermediate coupling is found to be long-range antiferromagnetic (AF) correlations [182]. The pseudogap disappears at large doping when the Fermi surface undergoes a topological transition from hole-like to electron-like. At strong coupling, the competition between various phases such as d -wave superconductivity, antiferromagnetic order and stripe states (spin/charge order modulating in real space) is an active subject of study. Recent studies using various methods like

DMRG and iPEPS found that the period 8 charge stripe state is lower in energy than the superconducting state at strong coupling $U/t = 8$ and no next-nearest-neighbour hopping [183]. This was later corroborated by studies from variational Monte Carlo and other DMRG studies [184, 185]. *It is not yet clear whether the stripe period gets lowered as the interaction strength is increased.* At weaker coupling, the above methods report a reduced stability for the strip state and increased tendency to realise a d -wave SC state [176, 186]. *Further investigation is necessary to map the crossovers between stripe and superconducting phases as doping is tuned. Moreover, the precise parameter regime in which the d -wave superconducting state becomes more stable than the stripe states is still an open question.*

At strong-coupling, the pseudogap is obtained below optimal doping and has been studied using dynamical methods such as cluster DMFT (DCA, dynamical cluster approximation) and cellular DMFT (CDMFT) [187–190]; in contrast to the weak-coupling feature (where the PG was borne from long-range AF fluctuations), it is believed that the momentum-space differentiation in this regime is brought about by the proximity to the Mott insulator [189]. Conflicting indications are however provided by a fluctuation analysis that finds that long-lived long-ranged spin-fluctuations at the antinode are responsible for the opening of the PG [191]. The PG is found to be accompanied by non-Fermi liquid behaviour in the one-particle self-energy [188]. The doping value at which the non-Fermi liquid first appears was found to be lower for electron-doping than hole-doping [192]. At half-filling, DCA calculations reveal that the extent of the PG decreases as the next-nearest-neighbour hopping strength is increased [193]. Moreover, while DCA+continuous-time auxiliary field methods do not show any effect of van Hove singularities on the generation of the PG [193], DCA+determinantal Monte Carlo approaches identify the proximity to van Hoves as favouring the formation of a PG [194]. *Methods with better momentum-space resolution is necessary in order to identify the role of nesting and van Hoves on the pseudogap.*

The question of whether superconductivity can arise from repulsion was one of the primary motivations for studying the Hubbard model. DiagMC calculations reveal superconductivity in the ground state at weak-coupling and moderate doping, and with multiple symmetries of the gap [176, 195]. For higher couplings (which is relevant to the cuprates) and moderate doping, stripe states take over. Quantum Monte Carlo and DCA calculations suggest a $d_{x^2-y^2}$ pairing symmetry [164, 196]. DCA calculations on the 2D Hubbard model suggest low-frequency spin fluctuations as the pairing glue for the superconductivity [197]. *The description of how various parent phases such as the pseudogap and strange metal precipitate superconductivity, in terms of the nature of fluctuations and the appropriate degrees of freedom, is still an open question..* Apart from superconductivity, several strongly correlated materials also exhibit strange metallicity, with transport properties that defy Fermi liquid theory, such as a linear-in-temperature resistivity that extends across large regimes of temperature. Within the Hubbard model, evidence for such T -linear resistivity and robustness across temperatures is available from CDMFT and determinantal Monte Carlo calculations [198, 199]. *More reliable methods are needed to access and understand the intermediate temperature regime of strange metals [141, 200].*

1.5.3 Theories of heavy fermion materials

Heavy fermion materials represent the canonical environment for studying quantum phase transitions in correlated systems, owing to the explicit observation of quantum critical points at the boundary of antiferromagnetic order [201]. These systems involve the coexistence of magnetic moments (arising from localised $4f$ - and $5f$ -electrons) and itinerant conduction electrons. The ground state of such a system is dictated by the competition between two kinds of interactions: internal spin-spin interaction J_{RKKY} between localised moments in the f -layer, and interaction J_K between f -electrons and conduction electrons [202].

Analogous to the screening of a single moment by a conduction bath, the lattice of f -moments are prone to undergoing Kondo screening by the conduction electrons through the inter-orbital hybridisation J_K . The local Fermi liquid excitations associated with the spins overlap with each other and become delocalised, leading to a paramagnetic heavy Fermi liquid with highly renormalised quasiparticle mass [203]. Because of the delocalisation of the excitations of the f -spin through Kondo screening, the number of carriers increases to $1 + x$ compared to x coming purely from the conduction electrons; this is commonly referred to as the large Fermi surface phase [204]. Indeed, a perturbation theoretic calculation in the on-site interaction strength of the f -layer allows one to rewrite the two-orbital model as two decoupled quasiparticle bands with dispersions renormalised by phenomenological parameters like the quasiparticle residue [74]. These phenomenological parameters can be estimated from mean-field approaches [205, 206]. While the temperature T^* at which coherent Landau quasiparticles can proliferate was found to be generically lower than the single impurity Kondo temperature T_K [207], T^* was found to be higher than T_K near half-filling of the conduction layer [208]. At half-filling in both layers, the chemical potential passes through the middle of the gap between the two quasiparticle bands, and the system is a Kondo insulator [209]; for bipartite lattices, the ground state can be generally shown to be unique and a singlet ($S = 0$) [210–212].

Tuning the f -layer J_{RKKY} to large values compared to the inter-layer hybridisation can bring forth a quantum phase transition into a different ground state [213]. Experimental evidence suggests that this transition usually falls into one of two classes, depending on the behaviour of the Kondo singlet binding energy E^* in the paramagnetic phase. If E^* remains non-zero as the QCP is approached from the heavy Fermi liquid into the ordered state, the transition is said to fall in the SDW type, because it can be described as an instability of the heavy quasiparticles into a spin-density-wave ordered phase [214–216], in the form of a quantum action that involves the spatial and temporal fluctuations of an order parameter. CeCu_2Si_2 is a prototypical material that exhibits quantum criticality of the SDW type [73]. Because of the non-zero binding energy of the singlet state at the QCP, heavy quasiparticles survive across the transition and the Fermi surface is expected to evolve continuously, along with the order parameter [217, 218].

On the other hand, if E^* vanishes at the QCP, the transition belongs to the Kondo breakdown type, which is not compatible with the spin-density-wave scenario [219]. This happens when gapless spin fluctuations within the f -layer compete with the Kondo spin-flip processes and lead to the destruction of Kondo screening on the Fermi surface [76]. This is tracked by the vanishing of the quasiparticle residue of the heavy Fermi liquid as one approaches the QCP from the paramagnetic phase [130, 218]; this in also implies the critical non-Fermi liquid nature of the large Fermi surface at the transition [220]. *A complete understanding of the Kondo breakdown class of QCP is still*

missing in the literature.

Due to the destruction of Kondo screening across the QCP in this class of materials, the localised f -spins cease to contribute to the Fermi surface away from the paramagnetic phase, and the Fermi surface shrinks discontinuously from large $(1+x)$ to small (x) across the QCP [221]. *Such a change in the Fermi volume violates Luttinger's theorem which equates the particles density to the Fermi surface volume, and Oshikawa's topological flux-tuning argument suggests the presence of additional degrees of freedom that respond to the flux while not being part of the Fermi surface. Additional work is needed to identify these degrees of freedom.* Materials like YbRh_2Si_2 and $\text{CuCu}_{6-x}\text{Au}_x$, among others, appear to exhibit behaviour that falls in this class [222, 223]. As mentioned before, several features of this class of experiments are not compatible with the SDW instability transition of the heavy Fermi liquid; these include the jump discontinuous change in Fermi volume mentioned previously (and an associated discontinuous change in the hall conductivity) [224], vanishing of the scale T_{FL} (associated with the onset of Fermi liquid behaviour) and the divergence of the quasiparticle mass upon approaching the QCP from the paramagnet [219], a universal linear-in-temperature strange metal behaviour of the resistivity in an intermediate finite temperature quantum critical regime above the QCP [222, 225] and the possible lack of magnetic ordering immediately away from the QCP [226, 227]. *At present, the connection of the physics of heavy fermion models like the periodic Anderson model to strange metal behaviour is an open question; the mechanism for generating such behaviour from the QCP remains to be completely understood.*

A fairly comprehensive model for describing the phase transition in heavy fermion systems is the periodic Anderson model (PAM):

$$H_{\text{PAM}} = \varepsilon_f \sum_{i,\sigma} n_{i,f,\sigma} + U_f \sum_i n_{i,f,\uparrow} n_{i,f,\downarrow} + \sum_{k,\sigma} \epsilon_k n_{k,c,\sigma} + V_{\perp} \sum_{i,\sigma} \left(c_{i,f,\sigma}^{\dagger} c_{i,c,\sigma} + \text{h.c.} \right), \quad (5.1)$$

where the subscripts f and c in the operators indicate whether they belong to the f -layer or the conduction electron layer. At the Kondo breakdown QCP and in the RKKY-dominated phase, the inter-layer hybridisation is expected to be irrelevant and the f - and conduction electrons fail to bind into a Kondo singlet in the ground state [228, 229]. Several numerical and analytical methods have been applied to investigate various parts of the phase diagram and probe varying aspects of the phenomenology. *However, many of these methods suffer from a lack of predictions of various transport measures like strange metallicity, and they often suffer from insufficient resolution in frequency or momentum space.*

One of the major approaches applied to this problem is that of extended DMFT (EMDFT), in which one maps the PAM to a self-consistently determined Bose-Fermi-Kondo model [130, 230]. One of the smoking gun signatures of quantum criticality is the scaling of dynamical properties as $\omega/k_B T$ due to temperature being the only remaining energy scale in the system; this is captured from EMDFT calculations [218]. Numerically exact unbiased methods like DMRG and auxiliary field quantum Monte Carlo demonstrate the abrupt change in Fermi surface volume across the transition [231, 232]. Parton methods that implement a slave-particle representation followed by a $1/N$ expansion reveal jump in the conductivity across the transition, with the small Fermi surface arising from a spin-charge separation of the f -electrons into a fractionalised Fermi liquid phase [233, 234].

Currently, a complete description of the Kondo breakdown driven transition that captures the behaviour of both the heavy Fermi liquid and the RKKY-dominated phase is missing.

1.6 Entanglement holography and fermionic criticality

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [235–240]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [241–261]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [237, 262, 263], topologically ordered systems have an additional subdominant topological term dubbed the topological entanglement entropy [264–282] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [239, 283, 284].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [285], appearing at the surfaces of three-dimensional topological insulators [286–289], quasi-2D organic materials [290–292] and in artificial materials like cold atoms, molecular and microwave crystals [293–299]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3} \ln l$ [300–309], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [307, 310, 311]. For massive fermions, the entanglement entropy (or rather the entropic c -function, which is the derivative of the entropy along the distance) can be expressed in the form of differential equations which need to be solved numerically [310, 312–314]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3} \ln \frac{l}{\epsilon} - \frac{1}{6} (ml \ln ml)^2 + O((ml)^2)$ for the case of small ml , and to $\sim (ml)^{-1/2} e^{-2ml}$ for that of large ml , both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1} \ln l$ violation of the area law, as observed in d -dimensional fermionic critical systems [310, 315–329]. This violation arises from the non-local nature of the quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [330, 331], it is possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [330, 332–335]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsys-

tem of sufficiently (as defined in Ref. [330]) large length is of the form $\frac{1}{6} \ln |2 \sin \frac{\phi}{2}|$ [330, 333]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [335–337]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [97–100, 338–340]. Given by the number of k -states enclosed by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger’s theorem [97–99]. As shown most prominently in Ref. [99], Luttinger’s theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger’s theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [100, 338–340]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Sec. ???. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [341–343], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [344–347]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [348–356], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [357–360]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [361–370]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [371–373]. The essence

of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [374–376] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of sub-system region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [377, 378]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [351, 352]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [379], the response functions of the Bose-Hubbard model [380, 381], and the phenomenology of non-Fermi liquids [382–385], holographic superconductors [386–391], striped phases [392–394] and holographic Fermi liquids [395–399]. An overview of the metallic states (phases that satisfy Luttinger’s theorem) and their connections to the gauge-gravity duality have been provided in Refs. [400–402].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [339, 340, 346, 347, 403–436]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [416–419] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [420–424]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving rise to the additional spatial dimension [350]. On the other hand, the MERA works by disentangling degrees of freedom in real space, creating a tensor network in the process [425–427] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [422]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [346, 347] (EHM) and the unitary renormalisation group [339, 340, 428] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson’s numerical renormalisation group [86, 429–431, 437], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [432–436].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [321, 323] as well as topological features arising from the incompressibility of the Fermi volume [99, 100]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [76, 339, 340]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important

systems such as unconventional superconductors, strange metals and heavy fermions [77, 438–440]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

Some open questions

1.7 A New Auxiliary Model Approach Towards Fermionic Criticality

To motivate the construction of a new auxiliary model approach in this thesis, we now write down some important outstanding questions that we will address using our approach.

1. *Can we construct an impurity-based auxiliary model approach that describes interaction-driven electronic phase transitions without imposing self-consistency?* As discussed in the preceding sections, one of the most prominent examples of auxiliary model approaches in the study of correlated fermionic models is dynamical mean-field theory (DMFT). However, solutions arising from DMFT and its extensions such as cellular DMFT and dynamical cluster approximation suffer from a lack of transparency owing to the numerical approach towards self-consistency. To go beyond such approaches, can we explicitly construct an impurity model that maps to the local physics of some interacting lattice model, and recover the low energy physics of the latter by solving the former? Apart from being much more transparent due to the explicit nature of the construction, such an approach would also be more tractable – because of the simplicity of impurity models – than studying the full lattice model.
2. *Can such an approach map the renormalisation group flows of an auxiliary impurity model to those of a correlated lattice model?* Both impurity and lattice models, due to the presence of many-particle interactions, typically involve a hierarchy of energy scales that are coupled to one another. The framework of renormalisation group (RG) is a powerful approach to identify the “important” physics at various energy scales. In particular, it can be used to identify low-energy effective Hamiltonians that describe the essential physics in the ground state and at low temperatures, and the microscopic Hamiltonian is said to “flow” into such low-energy fixed-point Hamiltonians. Can our auxiliary model approach map the renormalisation group flows between the impurity and lattice models, at least in terms of the local physics? This would make the underlying mechanism for the RG flows as well as the various fixed-point theories much more transparent.
3. *Does such an auxiliary model approach preserve symmetries of the microscopic/effective theories of the lattice model, and obey various conservation principles such as Luttinger’s theorem?* Importantly, an auxiliary model mapping for a lattice model should preserve the

symmetries of the full Hamiltonian. It should also, ideally, show the emergence of conservation principles and spectral sum rules. Luttinger’s theorem, as an example of such a sum rule, equates the electron density in a system with the Fermi volume in the presence of interactions, as long as the excitations are electronic. Violations in Luttinger’s theorem often signal interaction-driven transition from Fermi liquids to Mott insulators or non-Fermi liquids. Can the above auxiliary model approach capture changes in Luttinger’s theorem? An associated question is whether such an approach can show the emergence of Luttinger surfaces through the appearance of zeros in the one-particle Greens function. Such signatures would have topological consequences because the Luttinger sum rule is protected by topological invariants.

4. *Is it possible to describe topological transitions using such an approach?* It is becoming increasingly clear that various materials exhibit phase transitions that are not necessarily tied to the spontaneous breaking of a local symmetry and fall outside the Ginzburg-Landau-Wilson paradigm. Examples include (i) the Mott transition driven by a charge gap but without necessarily a spin gap, (ii) the Kondo breakdown QCP scenario in heavy fermion materials where the large-to-small Fermi surface transition is observed to be separate from magnetic ordering and (iii) transition from a trivial insulator to a topological insulator driven by changes in band topology. Such transitions do not exhibit the appearance of non-zero local order parameters, and are instead often described by changes in topological invariants and entanglement patterns. One can therefore ask: is such an auxiliary model approach capable of tracking such generalised order parameters?
5. *Can such an auxiliary model approach capture the effects of multiple orbitals and momentum-space differentiation?* One of the primary difficulties with many of the numerical methods typically employed in such problems is the high computational complexity (in time as well as memory) and the inability to resolve momentum-space features to sufficient precision. The increased complexity becomes particularly important in several classes of problems such as multi-band models, which are pertinent to investigations into orbital-selective transitions, heavy fermion physics and twisted bilayer graphene, among many others. The lack of momentum-space resolution is relevant to studies of the pseudogap (PG) phase that is observed in the underdoped high-T_c superconductors; the PG is known to display partial gapping of the Fermi surface around the antinodes while remaining gapless around the nodes. The approach proposed here can benefit from the explicit construction that avoids the computational cost of a self-consistency loop whilst providing access to a large number of momentum states, and the mapping to an impurity model which is much more simple to solve.

1.7.1 Important features of the “tiling” auxiliary model approach

Restoring translation symmetry using a many-body Blochs’ theorem

Mapping across Greens functions and entanglement measures

Local transition as a diagnostic of lattice transition

Lattice-embedding as a route to encoding lattice geometry

Bibliography

- [1] John Hubbard. Electron correlations in narrow energy bands. In *Proceedings of the royal society of london a: mathematical, physical and engineering sciences*, volume 276, pages 238–257. The Royal Society, 1963.
- [2] N. F. Mott. Metal-insulator transition. *Rev. Mod. Phys.*, 40:677–683, Oct 1968.
- [3] R F Milligan and G A Thomas. The metal-insulator transition. *Annual Review of Physical Chemistry*, 36(1):139–158, 1985.
- [4] Masatoshi Imada, Atsushi Fujimori, and Yoshinori Tokura. Metal-insulator transitions. *Reviews of modern physics*, 70(4):1039, 1998.
- [5] Patrick A. Lee, Naoto Nagaosa, and Xiao-Gang Wen. Doping a mott insulator: Physics of high-temperature superconductivity. *Rev. Mod. Phys.*, 78:17–85, Jan 2006.
- [6] A. Pustogow, M. Bories, A. Löhle, R. Rösslhuber, E. Zhukova, B. Gorshunov, S. Tomić, J. A. Schlueter, R. Hübner, T. Hiramatsu, Y. Yoshida, G. Saito, R. Kato, T.-H. Lee, V. Dobrosavljević, S. Fratini, and M. Dressel. Quantum spin liquids unveil the genuine mott state. *Nature Materials*, 17(9):773–777, Sep 2018.
- [7] G. R. Stewart. Non-fermi-liquid behavior in d - and f -electron metals. *Rev. Mod. Phys.*, 73:797–855, Oct 2001.
- [8] Yuan Cao, Valla Fatemi, Ahmet Demir, Shiang Fang, Spencer L. Tomarken, Jason Y. Luo, Javier D. Sanchez-Yamagishi, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, Ray C. Ashoori, and Pablo Jarillo-Herrero. Correlated insulator behaviour at half-filling in magic-angle graphene superlattices. *Nature*, 556(7699):80–84, Apr 2018.
- [9] Yuan Cao, Valla Fatemi, Shiang Fang, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, and Pablo Jarillo-Herrero. Unconventional superconductivity in magic-angle graphene superlattices. *Nature*, 556(7699):43–50, Apr 2018.
- [10] J H de Boer and E J W Verwey. Semi-conductors with partially and with completely filled 3d-lattice bands. *Proceedings of the Physical Society*, 49(4S):59, aug 1937.
- [11] N F Mott. The basis of the electron theory of metals, with special reference to the transition metals. *Proceedings of the Physical Society. Section A*, 62(7):416–422, jul 1949.

-
- [12] N. F. Mott. On the transition to metallic conduction in semiconductors. *Canadian Journal of Physics*, 34(12A):1356–1368, 1956.
 - [13] N. F. Mott. The transition to the metallic state. *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics*, 6(62):287–309, 1961.
 - [14] Takayuki Isono, Taichi Terashima, Kazuya Miyagawa, Kazushi Kanoda, and Shinya Uji. Quantum criticality in an organic spin-liquid insulator κ -(bedt-ttf) $_2$ Cu $_2$ (cn) $_3$. *Nature Communications*, 7(1):13494, Nov 2016.
 - [15] P. W. Anderson. New approach to the theory of superexchange interactions. *Phys. Rev.*, 115:2–13, Jul 1959.
 - [16] Junjiro Kanamori. Electron Correlation and Ferromagnetism of Transition Metals. *Progress of Theoretical Physics*, 30(3):275–289, 09 1963.
 - [17] Martin C. Gutzwiller. Effect of correlation on the ferromagnetism of transition metals. *Phys. Rev. Lett.*, 10:159–162, Mar 1963.
 - [18] Antoine Georges, Gabriel Kotliar, Werner Krauth, and Marcelo J Rozenberg. Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. *Reviews of Modern Physics*, 68(1):13, 1996.
 - [19] W. F. Brinkman and T. M. Rice. Application of gutzwiller’s variational method to the metal-insulator transition. *Phys. Rev. B*, 2:4302–4304, Nov 1970.
 - [20] Martin C. Gutzwiller. Correlation of electrons in a narrow s band. *Phys. Rev.*, 137:A1726–A1735, Mar 1965.
 - [21] Peter Prelovšek, Jure Kokalj, Zala Lenarčič, and Ross H. McKenzie. Holon-doublon binding as the mechanism for the mott transition. *Phys. Rev. B*, 92:235155, Dec 2015.
 - [22] Sen Zhou, Yupeng Wang, and Ziqiang Wang. Doublon-holon binding, mott transition, and fractionalized antiferromagnet in the hubbard model. *Phys. Rev. B*, 89:195119, May 2014.
 - [23] Tsutomu Watanabe, Hisatoshi Yokoyama, Yukio Tanaka, and Jun-ichiro Inoue. Superconductivity and a mott transition in a hubbard model on an anisotropic triangular lattice. *Journal of the Physical Society of Japan*, 75(7):074707, 2006.
 - [24] Philip Phillips. *Advanced solid state physics*. Cambridge University Press, 2012. Discussion regarding f-sum rule and dynamical spectral weight transfer in pages 394-395, section 12.4 on Cooper pairs and computation of effective two particle energy.
 - [25] G. Baskaran. An exactly solvable fermion model: spinons, holons and a non-fermi liquid phase. *Modern Physics Letters B*, 05(09):643–649, 1991.
 - [26] Yasuhiro Hatsugai and Mahito Kohmoto. Exactly solvable model of correlated lattice electrons in any dimensions. *Journal of the Physical Society of Japan*, 61(6):2056–2069, 1992.
-

- [27] Richard M. Martin, Lucia Reining, and David M. Ceperley. *Interacting Electrons: Theory and Computational Approaches*. Cambridge University Press, 2016.
- [28] Walter Metzner and Dieter Vollhardt. Correlated lattice fermions in $d = \infty$ dimensions. *Phys. Rev. Lett.*, 62:324–327, Jan 1989.
- [29] Antoine Georges. Strongly correlated electron materials: Dynamical mean-field theory and electronic structure. *AIP Conference Proceedings*, 715(1):3–74, 2004.
- [30] J. Zaanen, G. A. Sawatzky, and J. W. Allen. Band gaps and electronic structure of transition-metal compounds. *Phys. Rev. Lett.*, 55:418, 1985.
- [31] Masatoshi Imada, Atsushi Fujimori, and Yoshinori Tokura. Metal-insulator transitions. *Rev. Mod. Phys.*, 70:1039–1263, 1998.
- [32] D. B. McWhan, T. M. Rice, and J. P. Remeika. Mott transition in cr-doped v_2O_3 . *Phys. Rev. Lett.*, 23:1384–1387, Dec 1969.
- [33] P. Limelette, A. Georges, D. Jérôme, P. Wzietek, P. Metcalf, and J. M. Honig. Universality and critical behavior at the mott transition. *Science*, 302(5642):89–92, 2003.
- [34] M. J. Rozenberg, G. Kotliar, H. Kajueter, G. A. Thomas, D. H. Rapkine, J. M. Honig, and P. Metcalf. Optical conductivity in mott-hubbard systems. *Phys. Rev. Lett.*, 75:105–108, Jul 1995.
- [35] M. M. Qazilbash, M. Brehm, Byung-Gyu Chae, P.-C. Ho, G. O. Andreev, Bong-Jun Kim, Sun Jin Yun, A. V. Balatsky, M. B. Maple, F. Keilmann, Hyun-Tak Kim, and D. N. Basov. Mott transition in vo_2 revealed by infrared spectroscopy and nano-imaging. *Science*, 318(5857):1750–1753, 2007.
- [36] H. Takagi, B. Batlogg, H. L. Kao, J. Kwo, R. J. Cava, J. J. Krajewski, and W. F. Peck. Systematic evolution of temperature-dependent resistivity in $\text{la}_{2-x}\text{sr}_x\text{cuo}_4$. *Phys. Rev. Lett.*, 69:2975–2978, Nov 1992.
- [37] Yoichi Ando, Seiki Komiya, Kouji Segawa, S. Ono, and Y. Kurita. Electronic phase diagram of high- T_c cuprate superconductors from a mapping of the in-plane resistivity curvature. *Phys. Rev. Lett.*, 93:267001, Dec 2004.
- [38] T. R. Lemberger, I. Hetel, A. Tsukada, M. Naito, and M. Randeria. Superconductor-to-metal quantum phase transition in overdoped $\text{la}_{2-x}\text{sr}_x\text{cuo}_4$. *Phys. Rev. B*, 83:140507, Apr 2011.
- [39] Andrea Damascelli, Zahid Hussain, and Zhi-Xun Shen. Angle-resolved photoemission studies of the cuprate superconductors. *Rev. Mod. Phys.*, 75:473–541, Apr 2003.
- [40] Jonathan A. Sobota, Yu He, and Zhi-Xun Shen. Angle-resolved photoemission studies of quantum materials. *Rev. Mod. Phys.*, 93:025006, May 2021.

-
- [41] I M Vishik. Photoemission perspective on pseudogap, superconducting fluctuations, and charge order in cuprates: a review of recent progress. *Reports on Progress in Physics*, 81(6):062501, apr 2018.
- [42] María Luisa Medarde. Structural, magnetic and electronic properties of perovskites (r = rare earth). *Journal of Physics: Condensed Matter*, 9(8):1679, feb 1997.
- [43] Yuxin Wang, Kun Jiang, Jianjun Ying, Tao Wu, Jinguang Cheng, Jiangping Hu, and Xi-anhui Chen. Recent progress in nickelate superconductors. *National Science Review*, 12(10):nwaf373, 09 2025.
- [44] Pascal Puphal, Thomas Schäfer, Bernhard Keimer, and Matthias Hepting. Superconductivity in infinite-layer and ruddlesden–popper nickelates. *Nature Reviews Physics*, 8(2):70–85, Feb 2026.
- [45] Sajede Manzeli, Dmitry Ovchinnikov, Diego Pasquier, Oleg V. Yazyev, and Andras Kis. 2d transition metal dichalcogenides. *Nature Reviews Materials*, 2(8):17033, Jun 2017.
- [46] Saju Joseph, Jainy Mohan, Seetha Lakshmy, Simil Thomas, Brahmananda Chakraborty, Sabu Thomas, and Nandakumar Kalarikkal. A review of the synthesis, properties, and applications of 2d transition metal dichalcogenides and their heterostructures. *Materials Chemistry and Physics*, 297:127332, 2023.
- [47] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves in metallic, layered, transition-metal dichalcogenides. *Phys. Rev. Lett.*, 32:882–885, Apr 1974.
- [48] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves and superlattices in the metallic layered transition metal dichalcogenides. *Advances in Physics*, 50(8):1171–1248, 2001.
- [49] P. Fazekas and E. Tosatti. Electrical, structural and magnetic properties of pure and doped $1t\text{-tas}_2$. *Philosophical Magazine B*, 39:229–244, 1979.
- [50] L. Perfetti, T. A. Gloor, F. Mila, H. Berger, and M. Grioni. Unexpected periodicity in the quasi-two-dimensional mott insulator $1t\text{-tas}_2$ revealed by angle-resolved photoemission. *Phys. Rev. B*, 71:153101, Apr 2005.
- [51] B. Sipos, A. F. Kusmartseva, A. Akrap, H. Berger, L. Forró, and E. Tutiš. From mott state to superconductivity in $1t\text{-tas}_2$. *Nature Materials*, 7(12):960–965, Dec 2008.
- [52] Doohee Cho, Sangmo Cheon, Ki-Seok Kim, Sung-Hoon Lee, Yong-Heum Cho, Sang-Wook Cheong, and Han Woong Yeom. Nanoscale manipulation of the mott insulating state coupled to charge order in $1t\text{-tas}_2$. *Nature Communications*, 7(1):10453, Jan 2016.
- [53] S. V. Borisenko, A. A. Kordyuk, A. N. Yaresko, V. B. Zabolotnyy, D. S. Inosov, R. Schuster, B. Büchner, R. Weber, R. Follath, L. Patthey, and H. Berger. Pseudogap and charge density waves in two dimensions. *Phys. Rev. Lett.*, 100:196402, May 2008.
-

- [54] S. V. Borisenko, A. A. Kordyuk, V. B. Zabolotnyy, D. S. Inosov, D. Evtushinsky, B. Büchner, A. N. Yaresko, A. Varykhalov, R. Follath, W. Eberhardt, L. Patthey, and H. Berger. Two energy gaps and fermi-surface “arcs” in nbse₂. *Phys. Rev. Lett.*, 102:166402, Apr 2009.
- [55] V. Vescoli et al. Charge-density-wave correlations in the optical properties of layered transition-metal dichalcogenides. *Physical Review Letters*, 81:453–456, 1998.
- [56] Lijun Li, Xiaoyu Deng, Zhen Wang, Yu Liu, Milinda Abeykoon, Eric Dooryhee, Aleksandra Tomic, Yanan Huang, John B. Warren, Emil S. Bozin, Simon J. L. Billinge, Yuping Sun, Yimei Zhu, Gabriel Kotliar, and Cedomir Petrovic. Superconducting order from disorder in 2h-tase₂-xss. *npj Quantum Materials*, 2(1):11, Feb 2017.
- [57] Yoichi Kamihara, Takumi Watanabe, Masahiro Hirano, and Hideo Hosono. Iron-based layered superconductor la[o₁-xfx]feas (x = 0.05–0.12) with tc = 26 k. *Journal of the American Chemical Society*, 130(11):3296–3297, 2008. PMID: 18293989.
- [58] J. Paglione and R. L. Greene. High-temperature superconductivity in iron-based materials. *Nature Physics*, 6:645–658, 2010.
- [59] P. J. Hirschfeld, M. M. Korshunov, and I. I. Mazin. Gap symmetry and structure of fe-based superconductors. *Reports on Progress in Physics*, 74:124508, 2011.
- [60] G. R. Stewart. Superconductivity in iron compounds. *Rev. Mod. Phys.*, 83:1589–1652, Dec 2011.
- [61] David C. Johnston. The puzzle of high temperature superconductivity in layered iron pnictides and chalcogenides. *Advances in Physics*, 59(6):803–1061, 2010.
- [62] S. Kasahara, H. J. Shi, K. Hashimoto, S. Tonegawa, Y. Mizukami, T. Shibauchi, K. Sugimoto, T. Fukuda, T. Terashima, Andriy H. Nevidomskyy, and Y. Matsuda. Electronic nematicity above the structural and superconducting transition in bafe₂(as₁-x_p x)₂. *Nature*, 486(7403):382–385, Jun 2012.
- [63] R. M. Fernandes, A. V. Chubukov, and J. Schmalian. What drives nematic order in iron-based superconductors? *Nature Physics*, 10(2):97–104, Feb 2014.
- [64] James G. Analytis, H.-H. Kuo, Ross D. McDonald, Mark Wartenbe, P. M. C. Rourke, N. E. Hussey, and I. R. Fisher. Transport near a quantum critical point in bafe₂(as₁-x_px)₂. *Nature Physics*, 10(3):194–197, Mar 2014.
- [65] M. M. Qazilbash, J. J. Hamlin, R. E. Baumbach, Lijun Zhang, D. J. Singh, M. B. Maple, and D. N. Basov. Electronic correlations in the iron pnictides. *Nature Physics*, 5(9):647–650, Sep 2009.
- [66] H. Shishido, A. F. Bangura, A. I. Coldea, S. Tonegawa, K. Hashimoto, S. Kasahara, P. M. C. Rourke, H. Ikeda, T. Terashima, R. Settai, Y. Ōnuki, D. Vignolles, C. Proust, B. Vignolle, A. McCollam, Y. Matsuda, T. Shibauchi, and A. Carrington. Evolution of the fermi surface

- of $\text{BaFe}_2(\text{As}_{1-x}\text{P}_x)_2$ on entering the superconducting dome. *Phys. Rev. Lett.*, 104:057008, Feb 2010.
- [67] K. Terashima, Y. Sekiba, J. H. Bowen, K. Nakayama, T. Kawahara, T. Sato, P. Richard, Y.-M. Xu, L. J. Li, G. H. Cao, Z.-A. Xu, H. Ding, and T. Takahashi. Fermi surface nesting induced strong pairing in iron-based superconductors. *Proceedings of the National Academy of Sciences*, 106(18):7330–7333, 2009.
- [68] J.-H. Chu, H.-H. Kuo, J. G. Analytis, and I. R. Fisher. Divergent nematic susceptibility in an iron arsenide superconductor. *Science*, 337:710–712, 2012.
- [69] Hsueh-Hui Kuo, Jiun-Haw Chu, Johanna C. Palmstrom, Steven A. Kivelson, and Ian R. Fisher. Ubiquitous signatures of nematic quantum criticality in optimally doped Fe-based superconductors. *Science*, 352(6288):958–962, 2016.
- [70] D. S. Inosov, J. T. Park, P. Bourges, D. L. Sun, Y. Sidis, A. Schneidewind, K. Hradil, D. Haug, C. T. Lin, B. Keimer, and V. Hinkov. Normal-state spin dynamics and temperature-dependent spin-resonance energy in optimally doped $\text{BaFe}_{1.85}\text{Co}_{0.15}\text{As}_2$. *Nature Physics*, 6:178–181, 2010.
- [71] Andrey Chubukov. Pairing mechanism in Fe-based superconductors. *Annual Review of Condensed Matter Physics*, 3(Volume 3, 2012):57–92, 2012.
- [72] G. R. Stewart. Heavy-fermion systems. *Reviews of Modern Physics*, 56:755–787, 1984.
- [73] Steffen Wirth and Frank Steglich. Exploring heavy fermions from macroscopic to microscopic length scales. *Nature Reviews Materials*, 1(10):16051, Aug 2016.
- [74] A. C. Hewson. *The Kondo Problem to Heavy Fermions*. Cambridge University Press, 1993.
- [75] Hilbert v. Löhneysen, Achim Rosch, Matthias Vojta, and Peter Wölfle. Fermi-liquid instabilities at magnetic quantum phase transitions. *Rev. Mod. Phys.*, 79:1015–1075, Aug 2007.
- [76] Philipp Gegenwart, Qimiao Si, and Frank Steglich. Quantum criticality in heavy-fermion metals. *Nature Physics*, 4(3):186–197, Mar 2008.
- [77] J. Custers, P. Gegenwart, H. Wilhelm, K. Neumaier, Y. Tokiwa, O. Trovarelli, C. Geibel, F. Steglich, C. Pépin, and P. Coleman. The break-up of heavy electrons at a quantum critical point. *Nature*, 424(6948):524–527, Jul 2003.
- [78] S. Paschen, T. Lühmann, S. Wirth, P. Gegenwart, O. Trovarelli, C. Geibel, F. Steglich, P. Coleman, and Q. Si. Hall-effect evolution across a heavy-fermion quantum critical point. *Nature*, 432(7019):881–885, Dec 2004.
- [79] A. Schröder, G. Aeppli, R. Coldea, M. Adams, O. Stockert, H.v. Löhneysen, E. Bucher, R. Ramazashvili, and P. Coleman. Onset of antiferromagnetism in heavy-fermion metals. *Nature*, 407(6802):351–355, Sep 2000.

- [80] O. Stockert, J. Arndt, E. Faulhaber, C. Geibel, H. S. Jeevan, S. Kirchner, M. Loewenhaupt, K. Schmalzl, W. Schmidt, Q. Si, and F. Steglich. Magnetically driven superconductivity in CeCu_2Si_2 . *Nature Physics*, 7(2):119–124, Feb 2011.
- [81] Hiroaki Shishido, Rikio Settai, Hisatomo Harima, and Yoshichika Ōnuki. A drastic change of the fermi surface at a critical pressure in CeRhIn_5 : μSR study under pressure. *Journal of the Physical Society of Japan*, 74(4):1103–1106, 2005.
- [82] Qimiao Si and Frank Steglich. Heavy fermions and quantum phase transitions. *Science*, 329(5996):1161–1166, 2010.
- [83] Jun Kondo. Resistance minimum in dilute magnetic alloys. *Progress of theoretical physics*, 32(1):37–49, 1964.
- [84] JR Schrieffer and P.A.Wolff. Relation between the anderson and kondo hamiltonians. *Phys. Rev.*, 149:491, 1966.
- [85] Kenneth G Wilson. The renormalization group: Critical phenomena and the kondo problem. *Reviews of Modern Physics*, 47(4):773, 1975.
- [86] H. R. Krishna-murthy, J. W. Wilkins, and K. G. Wilson. Renormalization-group approach to the anderson model of dilute magnetic alloys. i. static properties for the symmetric case. *Phys. Rev. B*, 21:1003–1043, Feb 1980.
- [87] S Nishimoto and E Jeckelmann. Density-matrix renormalization group approach to quantum impurity problems. *Journal of Physics: Condensed Matter*, 16(4):613, jan 2004.
- [88] N. Andrei. Diagonalization of the kondo hamiltonian. *Phys. Rev. Lett.*, 45:379–382, Aug 1980.
- [89] P B Wiegmann. Exact solution of the s-d exchange model (kondo problem). *Journal of Physics C: Solid State Physics*, 14(10):1463–1478, apr 1981.
- [90] P Nozieres. A “fermi-liquid” description of the kondo problem at low temperatures. *Journal of Low Temperature Physics*, 17:31, 1974.
- [91] Ian Affleck. Conformal field theory approach to the kondo effect. *Acta Phys.Polon.B*, 26(1869), 1995.
- [92] T.A. Costi and A.C. Hewson. A new approach to the calculation of spectra for strongly correlated systems. *Physica B: Condensed Matter*, 163(1):179–181, 1990.
- [93] T.A. Costi and A.C. Hewson. Static and dynamic properties of the anderson model with conduction electron screening. *Physica C: Superconductivity*, 185-189:2649–2650, 1991.
- [94] Ryu Takayama and Osamu Sakai. Excitation spectra of the anderson model with coulomb interaction between localized and conduction electrons. *Physica B: Condensed Matter*, 186-188:915–917, 1993.

-
- [95] Wanda C. Oliveira and Luiz N. Oliveira. Generalized numerical renormalization-group method to calculate the thermodynamical properties of impurities in metals. *Phys. Rev. B*, 49:11986–11994, May 1994.
- [96] David C. Langreth. Friedel sum rule for anderson’s model of localized impurity states. *Phys. Rev.*, 150:516–518, Oct 1966.
- [97] JM Luttinger. Fermi surface and some simple equilibrium properties of a system of interacting fermions. *Physical Review*, 119(4):1153, 1960.
- [98] Joaquin Mazdak Luttinger and John Clive Ward. Ground-state energy of a many-fermion system. ii. *Physical Review*, 118(5):1417, 1960.
- [99] Masaki Oshikawa. Topological approach to luttinger’s theorem and the fermi surface of a kondo lattice. *Physical Review Letters*, 84(15):3370, 2000.
- [100] Kazuhiro Seki and Seiji Yunoki. Topological interpretation of the luttinger theorem. *Physical Review B*, 96(8):085124, 2017.
- [101] Nozières, Ph. and Blandin, A. Kondo effect in real metals. *J. Phys. France*, 41(3):”193–211”, 1980.
- [102] J Gan. On the multichannel kondo model. *Journal of Physics: Condensed Matter*, 6(24):4547, jun 1994.
- [103] Ian Affleck and Andreas W.W. Ludwig. Critical theory of overscreened kondo fixed points. *Nuclear Physics B*, 360(2):641–696, 1991.
- [104] Andreas W. W. Ludwig and Ian Affleck. Exact, asymptotic, three-dimensional, space- and time-dependent, green’s functions in the multichannel kondo effect. *Phys. Rev. Lett.*, 67:3160–3163, Nov 1991.
- [105] Ian Affleck, Andreas W. W. Ludwig, H.-B. Pang, and D. L. Cox. Relevance of anisotropy in the multichannel kondo effect: Comparison of conformal field theory and numerical renormalization-group results. *Phys. Rev. B*, 45:7918–7935, Apr 1992.
- [106] Ian Affleck and Andreas WW Ludwig. Exact conformal-field-theory results on the multichannel kondo effect: Single-fermion green’s function, self-energy, and resistivity. *Physical Review B*, 48(10):7297, 1993.
- [107] Olivier Parcollet and Antoine Georges. Transition from overscreening to underscreening in the multichannel kondo model: Exact solution at large N . *Phys. Rev. Lett.*, 79:4665–4668, Dec 1997.
- [108] Ian Affleck. Non-fermi liquid behavior in kondo models. *Journal of the Physical Society of Japan*, 74(1):59–66, 2005.
-

- [109] V. J. Emery and S. Kivelson. Mapping of the two-channel kondo problem to a resonant-level model. *Phys. Rev. B*, 46:10812–10817, Nov 1992.
- [110] David G. Clarke, Thierry Giamarchi, and Boris I. Shraiman. Curie and non-curie behavior of impurity spins in quantum antiferromagnets. *Phys. Rev. B*, 48:7070–7076, Sep 1993.
- [111] Gergely Zaránd and Jan von Delft. Analytical calculation of the finite-size crossover spectrum of the anisotropic two-channel kondo model. *Phys. Rev. B*, 61:6918–6933, Mar 2000.
- [112] Jan von Delft, Gergely Zaránd, and Michele Fabrizio. Finite-size bosonization of 2-channel kondo model: A bridge between numerical renormalization group and conformal field theory. *Phys. Rev. Lett.*, 81:196–199, Jul 1998.
- [113] A. J. Schofield. Bosonization in the two-channel kondo model. *Phys. Rev. B*, 55:5627–5630, Mar 1997.
- [114] R Bulla, T Costi, and T Pruschke. Numerical renormalisation group method for quantum impurity systems. *Rev. Mod. Phys.*, 80:395, 2008.
- [115] H. B. Pang and D. L. Cox. Stability of the fixed point of the two-channel kondo hamiltonian. *Phys. Rev. B*, 44:9454–9457, Nov 1991.
- [116] N. Andrei and C. Destri. Solution of the multichannel kondo problem. *Phys. Rev. Lett.*, 52:364–367, Jan 1984.
- [117] A. M. Tsvelick and P. B. Wiegmann. Solution of then-channel kondo problem (scaling and integrability). *Zeitschrift für Physik B Condensed Matter*, 54(3):201–206, Sep 1984.
- [118] A M Tsvelick. The thermodynamics of multichannel kondo problem. *Journal of Physics C: Solid State Physics*, 18(1):159–170, jan 1985.
- [119] Natan Andrei and Andrés Jerez. Fermi- and non-fermi-liquid behavior in the anisotropic multichannel kondo model: Bethe ansatz solution. *Phys. Rev. Lett.*, 74:4507–4510, May 1995.
- [120] Gergely Zaránd, Theo Costi, Andres Jerez, and Natan Andrei. Thermodynamics of the anisotropic two-channel kondo problem. *Phys. Rev. B*, 65:134416, Mar 2002.
- [121] A. V. Rozhkov. Impurity entropy for the two-channel kondo model. *International Journal of Modern Physics B*, 12(32):3457–3463, 1998.
- [122] Ru Zheng, Rong-Qiang He, and Zhong-Yi Lu. Order parameter for the multichannel kondo model at quantum criticality. *Phys. Rev. B*, 103:045111, Jan 2021.
- [123] D. Withoff and E. Fradkin. Phase transitions in gapless fermi systems with magnetic impurities. *Phys. Rev. Lett.*, 64:1835–1838, 1990.

-
- [124] C. Gonzalez-Buxton and K. Ingersent. Renormalization-group study of anderson and kondo impurities in gapless fermi systems. *Phys. Rev. B*, 57:14254–14269, 1998.
- [125] R. Bulla, T. Pruschke, and A. C. Hewson. Anderson impurity in pseudogap fermi systems. *J. Phys.: Condens. Matter*, 9:10463–10472, 1997.
- [126] K. Ingersent. Behavior of magnetic impurities in gapless fermi systems. *Phys. Rev. B*, 54:11936–11953, 1996.
- [127] K. Ingersent and Q. Si. Critical local-moment fluctuations in the pseudogap anderson model. *Phys. Rev. Lett.*, 89:076403, 2002.
- [128] L. Fritz and M. Vojta. Phase transitions in the pseudogap anderson and kondo models. *Phys. Rev. B*, 70:214427, 2004.
- [129] O. Parcollet and A. Georges. Non-fermi-liquid regime of a doped mott insulator. *Phys. Rev. B*, 59:5341–5360, 1999.
- [130] Qimiao Si, Silvio Rabello, Kevin Ingersent, and J. Llewellyn Smith. Locally critical quantum phase transitions in strongly correlated metals. *Nature*, 413(6858):804–808, Oct 2001.
- [131] R. Bulla, M. Vojta, and W. Hofstetter. Numerical renormalization group for quantum impurities in a pseudogap. *Phys. Rev. B*, 62:R6115–R6118, 2000.
- [132] B. A. Jones and C. M. Varma. Study of two magnetic impurities in a fermi gas. *Phys. Rev. Lett.*, 58:843–846, 1987.
- [133] B. A. Jones, C. M. Varma, and J. W. Wilkins. Low-temperature properties of the two-impurity kondo hamiltonian. *Phys. Rev. Lett.*, 61:125–128, 1988.
- [134] I. Affleck, A. W. W. Ludwig, and B. A. Jones. Conformal-field-theory approach to the two-impurity kondo problem. *Phys. Rev. B*, 52:9528–9546, 1995.
- [135] I. Affleck and A. W. W. Ludwig. Exact conformal-field-theory results on the multichannel kondo effect. *Phys. Rev. B*, 48:7297–7321, 1993.
- [136] Krzysztof P. Wójcik and Johann Kroha. Quantum spin liquid in an rkky-coupled two-impurity kondo system. *Phys. Rev. B*, 107:L121111, Mar 2023.
- [137] Elio J. König, Piers Coleman, and Yashar Komijani. Frustrated kondo impurity triangle: A simple model of deconfinement. *Phys. Rev. B*, 104:115103, Sep 2021.
- [138] A. Spinelli, M. Gerrits, R. Toskovic, B. Bryant, M. Ternes, and A. F. Otte. Exploring the phase diagram of the two-impurity kondo problem. *Nature Communications*, 6(1):10046, Nov 2015.
- [139] Hal Tasaki. The hubbard model - an introduction and selected rigorous results. *Journal of Physics: Condensed Matter*, 10(20):4353, may 1998.
-

- [140] Daniel P. Arovas, Erez Berg, Steven A. Kivelson, and Srinivas Raghu. The hubbard model. *Annual Review of Condensed Matter Physics*, 13(Volume 13, 2022):239–274, 2022.
- [141] Mingpu Qin, Thomas Schäfer, Sabine Andergassen, Philippe Corboz, and Emanuel Gull. The hubbard model: A computational perspective. *Annual Review of Condensed Matter Physics*, 13(Volume 13, 2022):275–302, 2022.
- [142] Elliott H Lieb and F Yu Wu. Absence of mott transition in an exact solution of the short-range, one-band model in one dimension. *Physical Review Letters*, 20(25):1445, 1968.
- [143] Antoine Georges and Gabriel Kotliar. Hubbard model in infinite dimensions. *Phys. Rev. B*, 45:6479–6483, Mar 1992.
- [144] V. Janis and D. Vollhardt. Comprehensive mean field theory for the hubbard model. *International Journal of Modern Physics B*, 06(05n06):731–747, 1992.
- [145] M. J. Rozenberg, X. Y. Zhang, and G. Kotliar. Mott-hubbard transition in infinite dimensions. *Phys. Rev. Lett.*, 69:1236–1239, Aug 1992.
- [146] Marcelo J. Rozenberg, R. Chitra, and Gabriel Kotliar. Finite temperature mott transition in the hubbard model in infinite dimensions. *Phys. Rev. Lett.*, 83:3498–3501, Oct 1999.
- [147] R. M. Noack and F. Gebhard. Mott-hubbard transition in infinite dimensions. *Phys. Rev. Lett.*, 82:1915–1918, Mar 1999.
- [148] P. Werner and A. J. Millis. Doping-driven mott transition in the one-band hubbard model. *Phys. Rev. B*, 75:085108, 2007.
- [149] Nozières, Ph. Some comments on kondo lattices and the mott transition. *Eur. Phys. J. B*, 6(4):”447–457”, ”” 1998.
- [150] G. Kotliar, Sahana Murthy, and M. J. Rozenberg. Compressibility divergence and the finite temperature mott transition. *Phys. Rev. Lett.*, 89:046401, Jul 2002.
- [151] Kotliar, G. Landau theory of the mott transition in the fully frustrated hubbard model in infinite dimensions. *Eur. Phys. J. B*, 11(1):27–39, 1999.
- [152] Xiaoyu Deng, Jernej Mravlje, Rok Žitko, Michel Ferrero, Gabriel Kotliar, and Antoine Georges. How bad metals turn good: Spectroscopic signatures of resilient quasiparticles. *Phys. Rev. Lett.*, 110:086401, Feb 2013.
- [153] R. Žitko, D. Hansen, E. Perepelitsky, J. Mravlje, A. Georges, and B. S. Shastry. Extremely correlated fermi liquid theory meets dynamical mean-field theory: Analytical insights into the doping-driven mott transition. *Phys. Rev. B*, 88:235132, Dec 2013.
- [154] J. Vučičević, D. Tanasković, M. J. Rozenberg, and V. Dobrosavljević. Bad-metal behavior reveals mott quantum criticality in doped hubbard models. *Phys. Rev. Lett.*, 114:246402, Jun 2015.

-
- [155] Nagamalleswararao Dasari, N. S. Vidhyadhiraja, Mark Jarrell, and Ross H. McKenzie. Quantum critical local spin dynamics near the mott metal-insulator transition in infinite dimensions. *Phys. Rev. B*, 95:165105, Apr 2017.
- [156] Heike Eisenlohr, Seung-Sup B. Lee, and Matthias Vojta. Mott quantum criticality in the one-band hubbard model: Dynamical mean-field theory, power-law spectra, and scaling. *Phys. Rev. B*, 100:155152, Oct 2019.
- [157] David E Logan and Martin R Galpin. Mott insulators and the doping-induced mott transition within DMFT: exact results for the one-band hubbard model. *Journal of Physics: Condensed Matter*, 28(2):025601, dec 2015.
- [158] Y. Ōno, R. Bulla, and A. C. Hewson. Phase diagram of the mott transition in a two-band hubbard model in infinite dimensions. *The European Physical Journal B - Condensed Matter and Complex Systems*, 19(3):375–384, Feb 2001.
- [159] Marcelo J Rozenberg. Integer-filling metal-insulator transitions in the degenerate hubbard model. *Physical Review B*, 55(8):R4855, 1997.
- [160] FC Zhang and TM Rice. Effective hamiltonian for the superconducting cu oxides. *Physical Review B*, 37(7):3759, 1988.
- [161] Motoharu Kitatani, Liang Si, Oleg Janson, Ryotaro Arita, Zhicheng Zhong, and Karsten Held. Nickelate superconductors - a renaissance of the one-band hubbard model. *npj Quantum Materials*, 5(1):59, Aug 2020.
- [162] B J Powell and Ross H McKenzie. Strong electronic correlations in superconducting organic charge transfer salts. *Journal of Physics: Condensed Matter*, 18(45):R827, oct 2006.
- [163] J. E. Hirsch. Two-dimensional hubbard model: Numerical simulation study. *Phys. Rev. B*, 31:4403–4419, Apr 1985.
- [164] Steven R White, Douglas J Scalapino, Robert L Sugar, EY Loh, James E Gubernatis, and Richard T Scalettar. Numerical study of the two-dimensional hubbard model. *Physical Review B*, 40(1):506, 1989.
- [165] T. Schäfer, F. Geles, D. Rost, G. Rohringer, E. Arrigoni, K. Held, N. Blümer, M. Aichhorn, and A. Toschi. Fate of the false mott-hubbard transition in two dimensions. *Phys. Rev. B*, 91:125109, Mar 2015.
- [166] Ettore Vitali, Hao Shi, Mingpu Qin, and Shiwei Zhang. Computation of dynamical correlation functions for many-fermion systems with auxiliary-field quantum monte carlo. *Phys. Rev. B*, 94:085140, Aug 2016.
- [167] Fedor Šimkovic, J. P. F. LeBlanc, Aaram J. Kim, Youjin Deng, N. V. Prokof'ev, B. V. Svistunov, and Evgeny Kozik. Extended crossover from a fermi liquid to a quasiantiferromagnet in the half-filled 2d hubbard model. *Phys. Rev. Lett.*, 124:017003, Jan 2020.
-

- [168] K. Borejsza and N. Dupuis. Antiferromagnetism and single-particle properties in the two-dimensional half-filled hubbard model: A nonlinear sigma model approach. *Phys. Rev. B*, 69:085119, Feb 2004.
- [169] Aaram J. Kim, Fedor Simkovic, and Evgeny Kozik. Spin and charge correlations across the metal-to-insulator crossover in the half-filled 2d hubbard model. *Phys. Rev. Lett.*, 124:117602, Mar 2020.
- [170] Marcel Klett, Nils Wentzell, Thomas Schäfer, Fedor Simkovic, Olivier Parcollet, Sabine Andergassen, and Philipp Hansmann. Real-space cluster dynamical mean-field theory: Center-focused extrapolation on the one- and two particle-levels. *Phys. Rev. Res.*, 2:033476, Sep 2020.
- [171] HJ Schulz. Incommensurate antiferromagnetism in the two-dimensional hubbard model. *Physical review letters*, 64(12):1445, 1990.
- [172] R. Frésard, M. Dzierzawa, and P. Wölfle. Slave-boson approach to spiral magnetic order in the hubbard model. *Europhysics Letters*, 15(3):325, jun 1991.
- [173] Walter Metzner, Manfred Salmhofer, Carsten Honerkamp, Volker Meden, and Kurt Schönhammer. Functional renormalization group approach to correlated fermion systems. *Reviews of Modern Physics*, 84(1):299, 2012.
- [174] Hiroyuki Yamase, Andreas Eberlein, and Walter Metzner. Coexistence of incommensurate magnetism and superconductivity in the two-dimensional hubbard model. *Physical review letters*, 116(9):096402, 2016.
- [175] S Raghu, SA Kivelson, and DJ Scalapino. Superconductivity in the repulsive hubbard model: An asymptotically exact weak-coupling solution. *Physical Review B*, 81(22):224505, 2010.
- [176] Fedor Šimkovic, Youjin Deng, and Evgeny Kozik. Superfluid ground state phase diagram of the two-dimensional hubbard model in the emergent bardeen-cooper-schrieffer regime. *Phys. Rev. B*, 104:L020507, Jul 2021.
- [177] Jing Wang, Andreas Eberlein, and Walter Metzner. Competing order in correlated electron systems made simple: Consistent fusion of functional renormalization and mean-field theory. *Phys. Rev. B*, 89:121116, Mar 2014.
- [178] M Zegrodnik, A Biborski, M Fidrysiak, and J Spałek. Superconductivity in the three-band model of cuprates: nodal direction characteristics and influence of intersite interactions. *Journal of Physics: Condensed Matter*, 33(41):415601, aug 2021.
- [179] C. Weber, C. Yee, K. Haule, and G. Kotliar. Scaling of the transition temperature of hole-doped cuprate superconductors with the charge-transfer energy. *Europhysics Letters*, 100(3):37001, nov 2012.

-
- [180] Steven R. White and D. J. Scalapino. Doping asymmetry and striping in a three-orbital CuO_2 hubbard model. *Phys. Rev. B*, 92:205112, Nov 2015.
- [181] Cornelia Hille, Daniel Rohe, Carsten Honerkamp, and Sabine Andergassen. Pseudogap opening in the two-dimensional hubbard model: a functional renormalization group analysis. *arXiv preprint arXiv:2003.01447*, 2020.
- [182] Wei Wu, Mathias S. Scheurer, Shubhayu Chatterjee, Subir Sachdev, Antoine Georges, and Michel Ferrero. Pseudogap and fermi-surface topology in the two-dimensional hubbard model. *Phys. Rev. X*, 8:021048, May 2018.
- [183] Bo-Xiao Zheng, Chia-Min Chung, Philippe Corboz, Georg Ehlers, Ming-Pu Qin, Reinhard M. Noack, Hao Shi, Steven R. White, Shiwei Zhang, and Garnet Kin-Lic Chan. Stripe order in the underdoped region of the two-dimensional hubbard model. *Science*, 358(6367):1155–1160, 2017.
- [184] Luca F. Tocchio, Arianna Montorsi, and Federico Becca. Metallic and insulating stripes and their relation with superconductivity in the doped Hubbard model. *SciPost Phys.*, 7:021, 2019.
- [185] Yi-Fan Jiang, Jan Zaanen, Thomas P. Devereaux, and Hong-Chen Jiang. Ground state phase diagram of the doped hubbard model on the four-leg cylinder. *Phys. Rev. Res.*, 2:033073, Jul 2020.
- [186] Mingpu Qin, Chia-Min Chung, Hao Shi, Ettore Vitali, Claudius Hubig, Ulrich Schollwöck, Steven R. White, and Shiwei Zhang. Absence of superconductivity in the pure two-dimensional hubbard model. *Phys. Rev. X*, 10:031016, Jul 2020.
- [187] Th. Maier, M. Jarrell, Th. Pruschke, and J. Keller. A non-crossing approximation for the study of intersite correlations. *The European Physical Journal B - Condensed Matter and Complex Systems*, 13(4):613–624, Feb 2000.
- [188] C. Huscroft, M. Jarrell, Th. Maier, S. Moukouri, and A. N. Tahvildarzadeh. Pseudogaps in the 2d hubbard model. *Phys. Rev. Lett.*, 86:139–142, Jan 2001.
- [189] M. Civelli, M. Capone, S. S. Kancharla, O. Parcollet, and G. Kotliar. Dynamical breakup of the fermi surface in a doped mott insulator. *Phys. Rev. Lett.*, 95:106402, Sep 2005.
- [190] O. Parcollet, G. Biroli, and G. Kotliar. Cluster dynamical mean field analysis of the mott transition. *Phys. Rev. Lett.*, 92:226402, Jun 2004.
- [191] O. Gunnarsson, T. Schäfer, J. P. F. LeBlanc, E. Gull, J. Merino, G. Sangiovanni, G. Rohringer, and A. Toschi. Fluctuation diagnostics of the electron self-energy: Origin of the pseudogap physics. *Phys. Rev. Lett.*, 114:236402, Jun 2015.
- [192] Ansgar Liebsch and Ning-Hua Tong. Finite-temperature exact diagonalization cluster dynamical mean-field study of the two-dimensional hubbard model: Pseudogap, non-fermi-liquid behavior, and particle-hole asymmetry. *Phys. Rev. B*, 80:165126, Oct 2009.
-

- [193] Emanuel Gull, Olivier Parcollet, Philipp Werner, and Andrew J. Millis. Momentum-sector-selective metal-insulator transition in the eight-site dynamical mean-field approximation to the hubbard model in two dimensions. *Phys. Rev. B*, 80:245102, Dec 2009.
- [194] Wei Wu, Mathias S. Scheurer, Michel Ferrero, and Antoine Georges. Effect of van hove singularities in the onset of pseudogap states in mott insulators. *Phys. Rev. Res.*, 2:033067, Jul 2020.
- [195] Youjin Deng, Evgeny Kozik, Nikolay V. Prokof'ev, and Boris V. Svistunov. Emergent bcs regime of the two-dimensional fermionic hubbard model: Ground-state phase diagram. *Europhysics Letters*, 110(5):57001, jun 2015.
- [196] Xi Chen, J. P. F. LeBlanc, and Emanuel Gull. Superconducting fluctuations in the normal state of the two-dimensional hubbard model. *Phys. Rev. Lett.*, 115:116402, Sep 2015.
- [197] E. Gull and A. J. Millis. Pairing glue in the two-dimensional hubbard model. *Phys. Rev. B*, 90:041110, Jul 2014.
- [198] Peter Cha, Aavishkar A. Patel, Emanuel Gull, and Eun-Ah Kim. Slope invariant t -linear resistivity from local self-energy. *Phys. Rev. Res.*, 2:033434, Sep 2020.
- [199] Edwin W. Huang, Ryan Sheppard, Brian Moritz, and Thomas P. Devereaux. Strange metallicity in the doped hubbard model. *Science*, 366(6468):987–990, 2019.
- [200] J. J. Vučićević, J. Kokalj, R. Žitko, N. Wentzell, D. Tanasković, and J. Mravlje. Conductivity in the square lattice hubbard model at high temperatures: Importance of vertex corrections. *Phys. Rev. Lett.*, 123:036601, Jul 2019.
- [201] Hilbert v. Löhneysen, Achim Rosch, Matthias Vojta, and Peter Wölfle. Fermi-liquid instabilities at magnetic quantum phase transitions. *Rev. Mod. Phys.*, 79:1015–1075, Aug 2007.
- [202] S. Doniach. The kondo lattice and weak antiferromagnetism. *Physica B+C*, 91:231–234, 1977.
- [203] Qimiao Si and Silke Paschen. Quantum phase transitions in heavy fermion metals and kondo insulators. *physica status solidi (b)*, 250(3):425–438, 2013.
- [204] Qimiao Si. Global magnetic phase diagram and local quantum criticality in heavy fermion metals. *Physica B: Condensed Matter*, 378-380:23–27, 2006. Proceedings of the International Conference on Strongly Correlated Electron Systems.
- [205] D.M. Newns and N. Read. Mean-field theory of intermediate valence/heavy fermion systems. *Advances in Physics*, 36(6):799–849, 1987.
- [206] A. Houghton, N. Read, and H. Won. Charge fluctuations, spin fluctuations, and superconductivity in the anderson lattice model of heavy-fermion systems. *Phys. Rev. B*, 37:3782–3785, Mar 1988.

-
- [207] S. Burdin, A. Georges, and D. R. Grempel. Coherence scale of the kondo lattice. *Phys. Rev. Lett.*, 85:1048–1051, Jul 2000.
- [208] Th. Pruschke, R. Bulla, and M. Jarrell. Low-energy scale of the periodic anderson model. *Phys. Rev. B*, 61:12799–12809, May 2000.
- [209] Hirokazu Tsunetsugu, Manfred Sigrist, and Kazuo Ueda. The ground-state phase diagram of the one-dimensional kondo lattice model. *Rev. Mod. Phys.*, 69:809–864, Jul 1997.
- [210] Kazuo Ueda, Hirokazu Tsunetsugu, and Manfred Sigrist. Singlet ground state of the periodic anderson model at half filling: A rigorous result. *Phys. Rev. Lett.*, 68:1030–1033, Feb 1992.
- [211] Shun-Qing Shen. Total spin and antiferromagnetic correlation in the kondo model. *Phys. Rev. B*, 53:14252–14261, Jun 1996.
- [212] Hirokazu Tsunetsugu. Rigorous results for half-filled kondo lattices. *Phys. Rev. B*, 55:3042–3045, Feb 1997.
- [213] Qimiao Si and Frank Steglich. Heavy fermions and quantum phase transitions. *Science*, 329(5996):1161–1166, 2010.
- [214] John A. Hertz. Quantum critical phenomena. *Phys. Rev. B*, 14:1165–1184, Aug 1976.
- [215] Tôru Moriya. *Spin Fluctuations in Itinerant Electron Magnetism*. Springer Berlin, Heidelberg, 1985.
- [216] A. J. Millis. Effect of a nonzero temperature on quantum critical points in itinerant fermion systems. *Phys. Rev. B*, 48:7183–7196, Sep 1993.
- [217] P Coleman, C Pépin, Qimiao Si, and R Ramazashvili. How do fermi liquids get heavy and die? *Journal of Physics: Condensed Matter*, 13(35):R723, aug 2001.
- [218] Qimiao Si, Silvio Rabello, Kevin Ingersent, and J. Llewellyn Smith. Local fluctuations in quantum critical metals. *Phys. Rev. B*, 68:115103, Sep 2003.
- [219] O. Stockert and F. Steglich. Unconventional quantum criticality in heavy-fermion compounds. *Annual Review of Condensed Matter Physics*, 2(Volume 2, 2011):79–99, 2011.
- [220] Peter Wölfle and Elihu Abrahams. Quasiparticles beyond the fermi liquid and heavy fermion criticality. *Phys. Rev. B*, 84:041101, Jul 2011.
- [221] Qimiao Si. Quantum criticality and global phase diagram of magnetic heavy fermions. *physica status solidi (b)*, 247(3):476–484, 2010.
- [222] O. Trovarelli, C. Geibel, S. Mederle, C. Langhammer, F. M. Grosche, P. Gegenwart, M. Lang, G. Sparn, and F. Steglich. YbRh_2Si_2 : Pronounced non-fermi-liquid effects above a low-lying magnetic phase transition. *Phys. Rev. Lett.*, 85:626–629, Jul 2000.
-

- [223] H.v Löhneysen, M Sieck, O Stockert, and M Waffenschmidt. Investigation of non-fermi-liquid behavior in $\text{CeCu}_6\text{-xAl}_x$. *Physica B: Condensed Matter*, 223-224:471–474, 1996. Proceedings of the International Conference on Strongly Correlated Electron Systems.
- [224] Sven Friedemann, Niels Oeschler, Steffen Wirth, Cornelius Krellner, Christoph Geibel, Frank Steglich, Silke Paschen, Stefan Kirchner, and Qimiao Si. Fermi-surface collapse and dynamical scaling near a quantum-critical point. *Proceedings of the National Academy of Sciences*, 107(33):14547–14551, 2010.
- [225] L. Prochaska, X. Li, D. C. MacFarland, A. M. Andrews, M. Bonta, E. F. Bianco, S. Yazdi, W. Schrenk, H. Detz, A. Limbeck, Q. Si, E. Ringe, G. Strasser, J. Kono, and S. Paschen. Singular charge fluctuations at a magnetic quantum critical point. *Science*, 367(6475):285–288, 2020.
- [226] S. Friedemann, T. Westerkamp, M. Brando, N. Oeschler, S. Wirth, P. Gegenwart, C. Krellner, C. Geibel, and F. Steglich. Detaching the antiferromagnetic quantum critical point from the fermi-surface reconstruction in YbRh_2Si_2 . *Nature Physics*, 5(7):465–469, Jul 2009.
- [227] Nikola Maksimovic, Daniel H. Eilbott, Tessa Cookmeyer, Fanghui Wan, Jan Ruzs, Vikram Nagarajan, Shannon C. Haley, Eran Maniv, Amanda Gong, Stefano Faubel, Ian M. Hayes, Ali Bangura, John Singleton, Johanna C. Palmstrom, Laurel Winter, Ross McDonald, Sooyoung Jang, Ping Ai, Yi Lin, Samuel Ciocys, Jacob Gobbo, Yochai Werman, Peter M. Oppeneer, Ehud Altman, Alessandra Lanzara, and James G. Analytis. Evidence for a delocalization quantum phase transition without symmetry breaking in CeCoIn_5 . *Science*, 375(6576):76–81, 2022.
- [228] Qimiao Si, Jedediah H. Pixley, Emilian Nica, Seiji J. Yamamoto, Pallab Goswami, Rong Yu, and Stefan Kirchner. Kondo destruction and quantum criticality in kondo lattice systems. *Journal of the Physical Society of Japan*, 83(6):061005, 2014.
- [229] Stefan Kirchner, Silke Paschen, Qiuyun Chen, Steffen Wirth, Donglai Feng, Joe D. Thompson, and Qimiao Si. Colloquium: Heavy-electron quantum criticality and single-particle spectroscopy. *Rev. Mod. Phys.*, 92:011002, Mar 2020.
- [230] Ping Sun and Gabriel Kotliar. Extended dynamical mean field theory study of the periodic anderson model. *Phys. Rev. Lett.*, 91:037209, Jul 2003.
- [231] Marcin Raczkowski, Bimla Danu, and Fakher F. Assaad. Breakdown of heavy quasiparticles in a honeycomb kondo lattice: A quantum monte carlo study. *Phys. Rev. B*, 106:L161115, Oct 2022.
- [232] Andreas Gleis, Jheng-Wei Li, and Jan von Delft. Controlled bond expansion for density matrix renormalization group ground state search at single-site costs. *Phys. Rev. Lett.*, 130:246402, Jun 2023.
- [233] T. Senthil, Matthias Vojta, and Subir Sachdev. Weak magnetism and non-fermi liquids near heavy-fermion critical points. *Phys. Rev. B*, 69:035111, Jan 2004.

-
- [234] P. Coleman, J. B. Marston, and A. J. Schofield. Transport anomalies in a simplified model for a heavy-electron quantum critical point. *Phys. Rev. B*, 72:245111, Dec 2005.
- [235] Barbara M. Terhal, Michael M. Wolf, and Andrew C. Doherty. Quantum entanglement: A modern perspective. *Physics Today*, 56(4):46–52, 2003.
- [236] Ryszard Horodecki, Paweł Horodecki, Michał Horodecki, and Karol Horodecki. Quantum entanglement. *Reviews of modern physics*, 81(2):865, 2009.
- [237] J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Rev. Mod. Phys.*, 82:277–306, Feb 2010.
- [238] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016.
- [239] Bei Zeng, Xie Chen, Duan-Lu Zhou, and Xiao-Gang Wen. *Quantum information meets quantum matter*. Springer, 2019.
- [240] Manuel Erhard, Mario Krenn, and Anton Zeilinger. Advances in high-dimensional quantum entanglement. *Nature Reviews Physics*, 2(7):365–381, Jul 2020.
- [241] X. G. Wen. Topological orders in rigid states. *International Journal of Modern Physics B*, 04(02):239–271, 1990.
- [242] X. G. Wen and Q. Niu. Ground-state degeneracy of the fractional quantum hall states in the presence of a random potential and on high-genus riemann surfaces. *Phys. Rev. B*, 41:9377–9396, May 1990.
- [243] X.G. Wen and A. Zee. Quantum statistics and superconductivity in two spatial dimensions. *Nuclear Physics B - Proceedings Supplements*, 15:135–156, 1990.
- [244] Xiao-Gang Wen. Topological orders and chern-simons theory in strongly correlated quantum liquid. *International Journal of Modern Physics B*, 05(10):1641–1648, 1991.
- [245] Xiao-Gang Wen. Topological orders and edge excitations in fractional quantum hall states. *Advances in Physics*, 44(5):405–473, 1995.
- [246] Xiao-Gang Wen. Topological order: From long-range entangled quantum matter to a unified origin of light and electrons. *ISRN Condensed Matter Physics*, 2013:198710, Mar 2013.
- [247] Xiao-Gang Wen. Colloquium: Zoo of quantum-topological phases of matter. *Reviews of Modern Physics*, 89(4):041004, 2017.
- [248] R. Tao and Yong-Shi Wu. Gauge invariance and fractional quantum hall effect. *Phys. Rev. B*, 30:1097–1098, Jul 1984.
- [249] Qian Niu, Ds J Thouless, and Yong-Shi Wu. Quantized hall conductance as a topological invariant. *Physical Review B*, 31(6):3372, 1985.
-

- [250] X. G. Wen. Vacuum degeneracy of chiral spin states in compactified space. *Phys. Rev. B*, 40:7387–7390, Oct 1989.
- [251] Hui Li and F Duncan M Haldane. Entanglement spectrum as a generalization of entanglement entropy: Identification of topological order in non-abelian fractional quantum hall effect states. *Physical review letters*, 101(1):010504, 2008.
- [252] Xiao-Liang Qi, Hosho Katsura, and Andreas W. W. Ludwig. General relationship between the entanglement spectrum and the edge state spectrum of topological quantum states. *Phys. Rev. Lett.*, 108:196402, May 2012.
- [253] Frank Pollmann, Ari M. Turner, Erez Berg, and Masaki Oshikawa. Entanglement spectrum of a topological phase in one dimension. *Phys. Rev. B*, 81:064439, Feb 2010.
- [254] Hong Yao and Xiao-Liang Qi. Entanglement entropy and entanglement spectrum of the kitaev model. *Phys. Rev. Lett.*, 105:080501, Aug 2010.
- [255] Zhao Liu, Hong-Li Guo, Vlatko Vedral, and Heng Fan. Entanglement spectrum: Identification of the transition from vortex-liquid to vortex-lattice state in a weakly interacting rotating bose-einstein condensate. *Phys. Rev. A*, 83:013620, Jan 2011.
- [256] Pasquale Calabrese and Alexandre Lefevre. Entanglement spectrum in one-dimensional systems. *Phys. Rev. A*, 78:032329, Sep 2008.
- [257] Andreas M. Läuchli, Emil J. Bergholtz, Juha Suorsa, and Masudul Haque. Disentangling entanglement spectra of fractional quantum hall states on torus geometries. *Phys. Rev. Lett.*, 104:156404, Apr 2010.
- [258] John Schliemann. Entanglement spectrum and entanglement thermodynamics of quantum hall bilayers at $\nu = 1$. *Phys. Rev. B*, 83:115322, Mar 2011.
- [259] B. I. Halperin. Quantized hall conductance, current-carrying edge states, and the existence of extended states in a two-dimensional disordered potential. *Phys. Rev. B*, 25:2185–2190, Feb 1982.
- [260] Xiao-Liang Qi, Hosho Katsura, and Andreas W. W. Ludwig. General relationship between the entanglement spectrum and the edge state spectrum of topological quantum states. *Phys. Rev. Lett.*, 108:196402, May 2012.
- [261] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016. Quantum entanglement in condensed matter systems.
- [262] Saurya Das and S. Shankaranarayanan. How robust is the entanglement entropy-area relation? *Phys. Rev. D*, 73:121701, Jun 2006.
- [263] M. Cramer, J. Eisert, M. B. Plenio, and J. Dreißig. Entanglement-area law for general bosonic harmonic lattice systems. *Phys. Rev. A*, 73:012309, Jan 2006.

-
- [264] B. Blok and X. G. Wen. Effective theories of the fractional quantum hall effect: Hierarchy construction. *Phys. Rev. B*, 42:8145–8156, Nov 1990.
- [265] N. Read. Excitation structure of the hierarchy scheme in the fractional quantum hall effect. *Phys. Rev. Lett.*, 65:1502–1505, Sep 1990.
- [266] Daniel S. Rokhsar and Steven A. Kivelson. Superconductivity and the quantum hard-core dimer gas. *Phys. Rev. Lett.*, 61:2376–2379, Nov 1988.
- [267] N. Read and B. Chakraborty. Statistics of the excitations of the resonating-valence-bond state. *Phys. Rev. B*, 40:7133–7140, Oct 1989.
- [268] R. Moessner and S. L. Sondhi. Resonating valence bond phase in the triangular lattice quantum dimer model. *Phys. Rev. Lett.*, 86:1881–1884, Feb 2001.
- [269] Eddy Ardonne, Paul Fendley, and Eduardo Fradkin. Topological order and conformal quantum critical points. *Annals of Physics*, 310(2):493–551, 2004.
- [270] Michael A. Levin and Xiao-Gang Wen. String-net condensation: A physical mechanism for topological phases. *Phys. Rev. B*, 71:045110, Jan 2005.
- [271] Alexei Kitaev and John Preskill. Topological entanglement entropy. *Physical review letters*, 96(11):110404, 2006.
- [272] Hong-Chen Jiang, Zhenghan Wang, and Leon Balents. Identifying topological order by entanglement entropy. *Nature Physics*, 8(12):902–905, Dec 2012.
- [273] Alioscia Hamma, Radu Ionicioiu, and Paolo Zanardi. Bipartite entanglement and entropic boundary law in lattice spin systems. *Phys. Rev. A*, 71:022315, Feb 2005.
- [274] Michael Levin and Xiao-Gang Wen. Detecting topological order in a ground state wave function. *Phys. Rev. Lett.*, 96:110405, Mar 2006.
- [275] Alexei Kitaev and John Preskill. Topological entanglement entropy. *Phys. Rev. Lett.*, 96:110404, Mar 2006.
- [276] Yi Zhang, Tarun Grover, Ari Turner, Masaki Oshikawa, and Ashvin Vishwanath. Quasiparticle statistics and braiding from ground-state entanglement. *Phys. Rev. B*, 85:235151, Jun 2012.
- [277] Tarun Grover, Ari M. Turner, and Ashvin Vishwanath. Entanglement entropy of gapped phases and topological order in three dimensions. *Phys. Rev. B*, 84:195120, Nov 2011.
- [278] Alioscia Hamma and Daniel A. Lidar. Adiabatic preparation of topological order. *Phys. Rev. Lett.*, 100:030502, Jan 2008.
- [279] Tian Lan and Xiao-Gang Wen. Topological quasiparticles and the holographic bulk-edge relation in $(2 + 1)$ -dimensional string-net models. *Phys. Rev. B*, 90:115119, Sep 2014.
-

- [280] Paul Fendley, Sergei V. Isakov, and Matthias Troyer. Fibonacci topological order from quantum nets. *Phys. Rev. Lett.*, 110:260408, Jun 2013.
- [281] Zheng-Cheng Gu, Michael Levin, Brian Swingle, and Xiao-Gang Wen. Tensor-product representations for string-net condensed states. *Phys. Rev. B*, 79:085118, Feb 2009.
- [282] Kevin Slagle, David Aasen, and Dominic Williamson. Foliated Field Theory and String-Membrane-Net Condensation Picture of Fracton Order. *SciPost Phys.*, 6:43, 2019.
- [283] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen. Local unitary transformation, long-range quantum entanglement, wave function renormalization, and topological order. *Physical review b*, 82(15):155138, 2010.
- [284] Zheng-Cheng Gu and Xiao-Gang Wen. Tensor-entanglement-filtering renormalization approach and symmetry-protected topological order. *Physical Review B*, 80(15):155131, 2009.
- [285] Oskar Vafek and Ashvin Vishwanath. Dirac fermions in solids: From high-tc cuprates and graphene to topological insulators and weyl semimetals. *Annual Review of Condensed Matter Physics*, 5(1):83–112, 2014.
- [286] B. Andrei Bernevig and Taylor L. Hughes. *Topological Insulators and Topological Superconductors*. Princeton University Press, 2013.
- [287] Jérôme Cayssol. Introduction to dirac materials and topological insulators. *Comptes Rendus Physique*, 14(9):760–778, 2013. Topological insulators / Isolants topologiques.
- [288] Roderich Moessner and Joel E. Moore. *Topological Phases of Matter*. Cambridge University Press, 2021.
- [289] Stephan Rachel. Interacting topological insulators: a review. *Reports on Progress in Physics*, 81(11):116501, oct 2018.
- [290] Jinying Wang, Shibin Deng, Zhongfan Liu, and Zhirong Liu. "The rare two-dimensional materials with Dirac cones". *National Science Review*, 2(1):22–39, 01 2015.
- [291] Menghao Wu, Zhijun Wang, Junwei Liu, Wenbin Li, Huahua Fu, Lei Sun, Xin Liu, Minghu Pan, Hongming Weng, Mircea Dincă, Liang Fu, and Ju Li. Conetronics in 2d metal-organic frameworks: double/half dirac cones and quantum anomalous hall effect. *2D Materials*, 4(1):015015, nov 2016.
- [292] Xiaojuan Ni, Huaqing Huang, and Jean-Luc Brédas. Emergence of a two-dimensional topological dirac semimetal phase in a phthalocyanine-based covalent organic framework. *Chemistry of Materials*, 34(7):3178–3184, 2022.
- [293] Shi-Liang Zhu, Baigeng Wang, and L.-M. Duan. Simulation and detection of dirac fermions with cold atoms in an optical lattice. *Phys. Rev. Lett.*, 98:260402, Jun 2007.

-
- [294] L. Lepori, G. Mussardo, and A. Trombettoni. (3+1) massive dirac fermions with ultracold atoms in frustrated cubic optical lattices. *EPL (Europhysics Letters)*, 92(5):50003, dec 2010.
- [295] O Boada, A Celi, J I Latorre, and M Lewenstein. Dirac equation for cold atoms in artificial curved spacetimes. *New Journal of Physics*, 13(3):035002, mar 2011.
- [296] Yoshihito Kuno, Ikuo Ichinose, and Yoshiro Takahashi. Generalized lattice wilson–dirac fermions in (1 + 1) dimensions for atomic quantum simulation and topological phases. *Scientific Reports*, 8(1):10699, Jul 2018.
- [297] Kenjiro K. Gomes, Warren Mar, Wonhee Ko, Francisco Guinea, and Hari C. Manoharan. Designer dirac fermions and topological phases in molecular graphene. *Nature*, 483(7389):306–310, Mar 2012.
- [298] Xiang Yuan, Cheng Zhang, Yanwen Liu, Awadhesh Narayan, Chaoyu Song, Shoudong Shen, Xing Sui, Jie Xu, Haochi Yu, Zhenghua An, Jun Zhao, Stefano Sanvito, Hugen Yan, and Faxian Xiu. Observation of quasi-two-dimensional dirac fermions in zrte5. *NPG Asia Materials*, 8(11):e325–e325, Nov 2016.
- [299] Matthieu Bellec, Ulrich Kuhl, Gilles Montambaux, and Fabrice Mortessagne. Topological transition of dirac points in a microwave experiment. *Phys. Rev. Lett.*, 110:033902, Jan 2013.
- [300] Guifre Vidal, José Ignacio Latorre, Enrique Rico, and Alexei Kitaev. Entanglement in quantum critical phenomena. *Physical review letters*, 90(22):227902, 2003.
- [301] J. I. Latorre, E. Rico, and G. Vidal. Ground state entanglement in quantum spin chains, 2003.
- [302] J. I. Latorre, C. A. Lütken, E. Rico, and G. Vidal. Fine-grained entanglement loss along renormalization-group flows. *Phys. Rev. A*, 71:034301, Mar 2005.
- [303] B.-Q. Jin and V. E. Korepin. Quantum spin chain, toeplitz determinants and the fisher—hartwig conjecture. *Journal of Statistical Physics*, 116(1):79–95, Aug 2004.
- [304] Neill Lambert, Clive Emary, and Tobias Brandes. Entanglement and the phase transition in single-mode superradiance. *Phys. Rev. Lett.*, 92:073602, Feb 2004.
- [305] Christoph Holzhey, Finn Larsen, and Frank Wilczek. Geometric and renormalized entropy in conformal field theory. *Nuclear Physics B*, 424(3):443–467, 1994.
- [306] H. Casini and M. Huerta. A finite entanglement entropy and the c-theorem. *Physics Letters B*, 600(1):142–150, 2004.
- [307] Pasquale Calabrese and John Cardy. Entanglement entropy and quantum field theory. *Journal of Statistical Mechanics: Theory and Experiment*, 2004(06):P06002, 2004.
- [308] Tobias J Osborne and Michael A Nielsen. Entanglement in a simple quantum phase transition. *Physical Review A*, 66(3):032110, 2002.
-

- [309] Andreas Osterloh, Luigi Amico, Giuseppe Falci, and Rosario Fazio. Scaling of entanglement close to a quantum phase transition. *Nature*, 416(6881):608–610, 2002.
- [310] H Casini, C D Fosco, and M Huerta. Entanglement and alpha entropies for a massive dirac field in two dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2005(07):P07007–P07007, jul 2005.
- [311] V. E. Korepin. Universality of entropy scaling in one dimensional gapless models. *Phys. Rev. Lett.*, 92:096402, Mar 2004.
- [312] J. L. Cardy, O. A. Castro-Alvaredo, and B. Doyon. Form factors of branch-point twist fields in quantum integrable models and entanglement entropy. *Journal of Statistical Physics*, 130(1):129–168, Jan 2008.
- [313] H Casini and M Huerta. Reduced density matrix and internal dynamics for multicomponent regions. *Classical and Quantum Gravity*, 26(18):185005, sep 2009.
- [314] Benjamin Doyon. Bipartite entanglement entropy in massive two-dimensional quantum field theory. *Phys. Rev. Lett.*, 102:031602, Jan 2009.
- [315] H Casini and M Huerta. Entanglement entropy in free quantum field theory. *Journal of Physics A: Mathematical and Theoretical*, 42(50):504007, dec 2009.
- [316] Dimitri Gioev and Israel Klich. Entanglement entropy of fermions in any dimension and the widom conjecture. *Physical review letters*, 96(10):100503, 2006.
- [317] Michael M. Wolf. Violation of the entropic area law for fermions. *Phys. Rev. Lett.*, 96:010404, Jan 2006.
- [318] Weifei Li, Letian Ding, Rong Yu, Tommaso Roscilde, and Stephan Haas. Scaling behavior of entanglement in two- and three-dimensional free-fermion systems. *Phys. Rev. B*, 74:073103, Aug 2006.
- [319] T. Barthel, M.-C. Chung, and U. Schollwöck. Entanglement scaling in critical two-dimensional fermionic and bosonic systems. *Phys. Rev. A*, 74:022329, Aug 2006.
- [320] S. Farkas and Z. Zimborás. The von neumann entropy asymptotics in multidimensional fermionic systems. *Journal of Mathematical Physics*, 48(10):102110, 2007.
- [321] Michael M Wolf, Frank Verstraete, Matthew B Hastings, and J Ignacio Cirac. Area laws in quantum systems: mutual information and correlations. *Physical review letters*, 100(7):070502, 2008.
- [322] Olalla A Castro-Alvaredo and Benjamin Doyon. Bi-partite entanglement entropy in massive (1+1)-dimensional quantum field theories. *Journal of Physics A: Mathematical and Theoretical*, 42(50):504006, dec 2009.

-
- [323] Brian Swingle. Entanglement entropy and the fermi surface. *Physical review letters*, 105(5):050502, 2010.
 - [324] Robert Helling, Hajo Leschke, and Wolfgang Spitzer. "A Special Case of a Conjecture by Widom with Implications to Fermionic Entanglement Entropy". *International Mathematics Research Notices*, 2011(7):1451–1482, 06 2010.
 - [325] P. Calabrese, M. Mintchev, and E. Vicari. Entanglement entropies in free-fermion gases for arbitrary dimension. *EPL (Europhysics Letters)*, 97(2):20009, jan 2012.
 - [326] Brian Swingle. Rényi entropy, mutual information, and fluctuation properties of fermi liquids. *Phys. Rev. B*, 86:045109, Jul 2012.
 - [327] Pasquale Calabrese, Mihail Mintchev, and Ettore Vicari. Exact relations between particle fluctuations and entanglement in fermi gases. *EPL (Europhysics Letters)*, 98(2):20003, apr 2012.
 - [328] Mao Tian Tan and Shinsei Ryu. Particle number fluctuations, renyi entropy, and symmetry-resolved entanglement entropy in a two-dimensional fermi gas from multidimensional bosonization. *Phys. Rev. B*, 101:235169, Jun 2020.
 - [329] Sara Murciano, Paola Ruggiero, and Pasquale Calabrese. Symmetry resolved entanglement in two-dimensional systems via dimensional reduction. *Journal of Statistical Mechanics: Theory and Experiment*, 2020(8):083102, aug 2020.
 - [330] Xiao Chen, William Witczak-Krempa, Thomas Faulkner, and Eduardo Fradkin. Two-cylinder entanglement entropy under a twist. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(4):043104, apr 2017.
 - [331] Pablo Bueno and William Witczak-Krempa. Holographic torus entanglement and its renormalization group flow. *Phys. Rev. D*, 95:066007, Mar 2017.
 - [332] Max A. Metlitski, Carlos A. Fuertes, and Subir Sachdev. Entanglement entropy in the $o(n)$ model. *Phys. Rev. B*, 80:115122, Sep 2009.
 - [333] Raúl E Arias, David D Blanco, and Horacio Casini. Entanglement entropy as a witness of the aharonov–bohm effect in QFT. *Journal of Physics A: Mathematical and Theoretical*, 48(14):145401, mar 2015.
 - [334] Seth Whitsitt, William Witczak-Krempa, and Subir Sachdev. Entanglement entropy of large- n wilson-fisher conformal field theory. *Phys. Rev. B*, 95:045148, Jan 2017.
 - [335] Wei Zhu, Xiao Chen, Yin-Chen He, and William Witczak-Krempa. Entanglement signatures of emergent dirac fermions: Kagome spin liquid and quantum criticality. *Science Advances*, 4(11):eaat5535, 2018.
 - [336] Shijie Hu, W. Zhu, Sebastian Eggert, and Yin-Chen He. Dirac spin liquid on the spin-1/2 triangular heisenberg antiferromagnet. *Phys. Rev. Lett.*, 123:207203, Nov 2019.
-

- [337] Valentin Crépel, Anna Hackenbroich, Nicolas Regnault, and Benoit Estienne. Universal signatures of dirac fermions in entanglement and charge fluctuations. *Phys. Rev. B*, 103:235108, Jun 2021.
- [338] Joshua T Heath and Kevin S Bedell. Necessary and sufficient conditions for the validity of luttinger’s theorem. *New Journal of Physics*, 22(6):063011, jun 2020.
- [339] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-i: a tensor network approach. *Nuclear Physics B*, 960:115170, 2020.
- [340] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-ii: insights on fermionic criticality. *Nuclear Physics B*, 960:115163, 2020.
- [341] G. ’t Hooft. Dimensional reduction in quantum gravity. *arXiv:gr-qc/9310026*, 1993.
- [342] Erik Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011(4):29, Apr 2011.
- [343] Hsu-Wen Chiang, Yao-Chieh Hu, and Pisin Chen. Quantization of spacetime based on a spacetime interval operator. *Phys. Rev. D*, 93:084043, Apr 2016.
- [344] Mark Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42(10):2323–2329, 2010.
- [345] Mark Van Raamsdonk. *Lectures on Gravity and Entanglement*, chapter Chapter 5, pages 297–351. World Scientific, 2017.
- [346] Xiao-Liang Qi. Exact holographic mapping and emergent space-time geometry. *arXiv preprint arXiv:1309.6282*, 2013.
- [347] Ching Hua Lee and Xiao-Liang Qi. Exact holographic mapping in free fermion systems. *Physical Review B*, 93(3):035112, 2016.
- [348] Raphael Bousso. The holographic principle. *Rev. Mod. Phys.*, 74:825–874, Aug 2002.
- [349] Joseph Katz and Jian Sheng. Applications of holography in fluid mechanics and particle dynamics. *Annual Review of Fluid Mechanics*, 42(1):531–555, 2010.
- [350] A.G. Green. An introduction to gauge-gravity duality and its application in condensed matter. *Contemporary Physics*, 54(1):33–48, 2013.
- [351] Sean A Hartnoll. Lectures on holographic methods for condensed matter physics. *Classical and Quantum Gravity*, 26(22):224002, oct 2009.
- [352] Subir Sachdev. What can gauge-gravity duality teach us about condensed matter physics? *Annual Review of Condensed Matter Physics*, 3(1):9–33, 2012.
- [353] Jan Zaanen, Yan Liu, Ya-Wen Sun, and Koenraad Schalm. *Holographic Duality in Condensed Matter Physics*. Cambridge University Press, 2015.

-
- [354] Anosh Joseph. Review of lattice supersymmetry and gauge-gravity duality. *International Journal of Modern Physics A*, 30(27):1530054, 2015.
- [355] Horațiu Năstase. *Introduction to the AdS/CFT Correspondence*. Cambridge University Press, 2015.
- [356] Johanna Erdmenger. Introduction to gauge/gravity duality (tasi lectures 2017), 2018.
- [357] G. 't Hooft. A planar diagram theory for strong interactions. *Nuclear Physics B*, 72(3):461–473, 1974.
- [358] A.M. Polyakov. Thermal properties of gauge fields and quark liberation. *Physics Letters B*, 72(4):477–480, 1978.
- [359] A.M. Polyakov. *Gauge Fields and Strings (1st ed.)*. Routledge, 1987.
- [360] David J. Gross and Washington Taylor. Two-dimensional qcd is a string theory. *Nuclear Physics B*, 400(1):181–208, 1993.
- [361] E.T. Akhmedov. A remark on the ads/cft correspondence and the renormalization group flow. *Physics Letters B*, 442(1):152–158, 1998.
- [362] Enrique Álvarez and César Gómez. Geometric holography, the renormalization group and the c-theorem. *Nuclear Physics B*, 541(1):441–460, 1999.
- [363] D. Z. Freedman, S. S. Gubser, K. Pilch, and N. P. Warner. Renormalization group flows from holography–supersymmetry and a c-theorem. *Advances in Theoretical and Mathematical Physics*, 3(2):363–417, 1999.
- [364] M. Porrati and A. Starinets. Rg fixed points in supergravity duals of 4-d field theory and asymptotically ads spaces. *Physics Letters B*, 454(1):77–83, 1999.
- [365] Shamit Kachru, Michael Schulz, and Eva Silverstein. Self-tuning flat domain walls in 5d gravity and string theory. *Phys. Rev. D*, 62:045021, Jul 2000.
- [366] L. Girardello, M. Petrini, M. Porrati, and A. Zaffaroni. The supergravity dual of $n=1$ super yang–mills theory. *Nuclear Physics B*, 569(1):451–469, 2000.
- [367] Jan De Boer, Erik Verlinde, and Herman Verlinde. On the holographic renormalization group. *Journal of High Energy Physics*, 2000(08):003, 2000.
- [368] Jan de Boer. The holographic renormalization group. *Fortschritte der Physik*, 49(4-6):339–358, 2001.
- [369] Clifford V Johnson, Kenneth J Lovis, and David C Page. Probing some $N = 1$ AdS/CFT RG flows. *Journal of High Energy Physics*, 2001(05):036–036, may 2001.
- [370] Satoshi Yamaguchi. AdS branes corresponding to superconformal defects. *Journal of High Energy Physics*, 2003(06):002–002, jun 2003.
-

- [371] Juan Maldacena. The large- n limit of superconformal field theories and supergravity. *International journal of theoretical physics*, 38(4):1113–1133, 1999.
- [372] S.S. Gubser, I.R. Klebanov, and A.M. Polyakov. Gauge theory correlators from non-critical string theory. *Physics Letters B*, 428(1):105–114, 1998.
- [373] Edward Witten. Anti de sitter space and holography. *arXiv preprint hep-th/9802150*, 2:253–291, 1998.
- [374] Jacob D. Bekenstein. Black holes and entropy. *Phys. Rev. D*, 7:2333–2346, Apr 1973.
- [375] S. W. Hawking. Black hole explosions? *Nature*, 248(5443):30–31, Mar 1974.
- [376] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43(3):199–220, Aug 1975.
- [377] Shinsei Ryu and Tadashi Takayanagi. Holographic derivation of entanglement entropy from the anti-de sitter space/conformal field theory correspondence. *Physical review letters*, 96(18):181602, 2006.
- [378] Shinsei Ryu and Tadashi Takayanagi. Aspects of holographic entanglement entropy. *Journal of High Energy Physics*, 2006(08):045, 2006.
- [379] P. K. Kovtun, D. T. Son, and A. O. Starinets. Viscosity in strongly interacting quantum field theories from black hole physics. *Phys. Rev. Lett.*, 94:111601, Mar 2005.
- [380] Kedar Damle and Subir Sachdev. Nonzero-temperature transport near quantum critical points. *Phys. Rev. B*, 56:8714–8733, Oct 1997.
- [381] Sean A. Hartnoll, Pavel K. Kovtun, Markus Müller, and Subir Sachdev. Theory of the nernst effect near quantum phase transitions in condensed matter and in dyonic black holes. *Phys. Rev. B*, 76:144502, Oct 2007.
- [382] Sung-Sik Lee. Non-fermi liquid from a charged black hole: A critical fermi ball. *Phys. Rev. D*, 79:086006, Apr 2009.
- [383] Hong Liu, John McGreevy, and David Vegh. Non-fermi liquids from holography. *Phys. Rev. D*, 83:065029, Mar 2011.
- [384] Thomas Faulkner, Hong Liu, John McGreevy, and David Vegh. Emergent quantum criticality, fermi surfaces, and ads_2 . *Phys. Rev. D*, 83:125002, Jun 2011.
- [385] Mihailo Čubrović, Jan Zaanen, and Koenraad Schalm. String theory, quantum phase transitions, and the emergent fermi liquid. *Science*, 325(5939):439–444, 2009.
- [386] Steven S. Gubser, Silviu S. Pufu, and Fabio D. Rocha. Quantum critical superconductors in string theory and m-theory. *Physics Letters B*, 683(2):201–204, 2010.

-
- [387] Jerome P. Gauntlett, Julian Sonner, and Toby Wiseman. Holographic superconductivity in m theory. *Phys. Rev. Lett.*, 103:151601, Oct 2009.
- [388] Sean A. Hartnoll, Christopher P. Herzog, and Gary T. Horowitz. Building a holographic superconductor. *Phys. Rev. Lett.*, 101:031601, Jul 2008.
- [389] Sean A Hartnoll, Christopher P Herzog, and Gary T Horowitz. Holographic superconductors. *Journal of High Energy Physics*, 2008(12):015–015, dec 2008.
- [390] RongGen Cai, Li Li, LiFang Li, and RunQiu Yang. Introduction to holographic superconductor models. *Science China Physics, Mechanics & Astronomy*, 58(6):1–46, Jun 2015.
- [391] Chunyan Wang, Dan Zhang, Guoyang Fu, and Jian-Pin Wu. Analytical study of the holographic superconductor from higher derivative theory. *Advances in High Energy Physics*, 2020:5902473, Jan 2020.
- [392] Aristomenis Donos and Jerome P. Gauntlett. Holographic striped phases. *Journal of High Energy Physics*, 2011(8):140, Aug 2011.
- [393] Aristomenis Donos. Striped phases from holography. *Journal of High Energy Physics*, 2013(5):59, May 2013.
- [394] Sera Cremonini, Li Li, and Jie Ren. Holographic fermions in striped phases. *Journal of High Energy Physics*, 2018(12):80, Dec 2018.
- [395] Shamit Kachru, Xiao Liu, and Michael Mulligan. Gravity duals of lifshitz-like fixed points. *Phys. Rev. D*, 78:106005, Nov 2008.
- [396] Mihailo Čubrović, Yan Liu, Koenraad Schalm, Ya-Wen Sun, and Jan Zaanen. Spectral probes of the holographic fermi ground state: Dialing between the electron star and ads dirac hair. *Phys. Rev. D*, 84:086002, Oct 2011.
- [397] T. Senthil, Subir Sachdev, and Matthias Vojta. Fractionalized fermi liquids. *Phys. Rev. Lett.*, 90:216403, May 2003.
- [398] Subir Sachdev. Model of a fermi liquid using gauge-gravity duality. *Phys. Rev. D*, 84:066009, Sep 2011.
- [399] Sean A. ”Hartnoll, Diego M. Hofman, and David” Vegh. ”Stellar spectroscopy: Fermions and holographic Lifshitz criticality”. *”JHEP”*, ”08”.”096”, ”2011”.
- [400] Liza Huijse and Subir Sachdev. Fermi surfaces and gauge-gravity duality. *Phys. Rev. D*, 84:026001, Jul 2011.
- [401] Francesco Aprile, Diederik Roest, and Jorge G. Russo. Holographic superconductors from gauged supergravity. *Journal of High Energy Physics*, 2011(6):40, Jun 2011.
-

- [402] Aristomenis Donos and Jerome P. Gauntlett. Superfluid black branes in $ads_4 \times s_7$. *Journal of High Energy Physics*, 2011(6):53, Jun 2011.
- [403] Hiroaki Matsueda. Multiscale entanglement renormalization ansatz for kondo problem. *arXiv:1208.2872*, 2012.
- [404] Hiroaki Matsueda, Masafumi Ishihara, and Yoichiro Hashizume. Tensor network and a black hole. *Phys. Rev. D*, 87:066002, Mar 2013.
- [405] Hiroaki Matsueda. Emergent general relativity from fisher information metric. *arXiv:1310.1831*, 2013.
- [406] Hiroaki Matsueda. Geometry and dynamics of emergent spacetime from entanglement spectrum, 2014.
- [407] Bartłomiej Czech. Einstein equations from varying complexity. *Phys. Rev. Lett.*, 120:031601, Jan 2018.
- [408] Thomas Faulkner, Felix M. Haehl, Eliot Hijano, Onkar Parrikar, Charles Rabideau, and Mark Van Raamsdonk. Nonlinear gravity from entanglement in conformal field theories. *Journal of High Energy Physics*, 2017(8):57, Aug 2017.
- [409] Thomas Faulkner, Monica Guica, Thomas Hartman, Robert C. Myers, and Mark Van Raamsdonk. Gravitation from entanglement in holographic cfts. *Journal of High Energy Physics*, 2014(3):51, Mar 2014.
- [410] Yingfei Gu, Ching Hua Lee, Xueda Wen, Gil Young Cho, Shinsei Ryu, and Xiao-Liang Qi. Holographic duality between $(2+1)$ -dimensional quantum anomalous hall state and $(3+1)$ -dimensional topological insulators. *Phys. Rev. B*, 94:125107, Sep 2016.
- [411] Xueda Wen, Gil Young Cho, Pedro L. S. Lopes, Yingfei Gu, Xiao-Liang Qi, and Shinsei Ryu. Holographic entanglement renormalization of topological insulators. *Phys. Rev. B*, 94:075124, Aug 2016.
- [412] Robert G. Leigh, Onkar Parrikar, and Alexander B. Weiss. Exact renormalization group and higher-spin holography. *Phys. Rev. D*, 91:026002, Jan 2015.
- [413] Robert G. Leigh, Onkar Parrikar, and Alexander B. Weiss. Holographic geometry of the renormalization group and higher spin symmetries. *Phys. Rev. D*, 89:106012, May 2014.
- [414] Michael R. Douglas, Luca Mazzucato, and Shlomo S. Razamat. Holographic dual of free field theory. *Phys. Rev. D*, 83:071701, Apr 2011.
- [415] Robert de Mello Koch, Antal Jevicki, Kewang Jin, and João P. Rodrigues. ads_4/cft_3 construction from collective fields. *Phys. Rev. D*, 83:025006, Jan 2011.
- [416] Sung-Sik Lee. Holographic description of quantum field theory. *Nuclear Physics B*, 832(3):567–585, 2010.

-
- [417] Sung-Sik Lee. Tasi lectures on emergence of supersymmetry, gauge theory and string in condensed matter systems. *arXiv:1009.5127*, 2010.
- [418] Sung-Sik Lee. Background independent holographic description: from matrix field theory to quantum gravity. *Journal of High Energy Physics*, 2012(10):160, Oct 2012.
- [419] Sung-Sik Lee. Quantum renormalization group and holography. *Journal of High Energy Physics*, 2014(1):76, 2014.
- [420] Guifre Vidal. Entanglement renormalization. *Physical review letters*, 99(22):220405, 2007.
- [421] Guifré Vidal. Class of quantum many-body states that can be efficiently simulated. *Physical review letters*, 101(11):110501, 2008.
- [422] Brian Swingle. Entanglement renormalization and holography. *Physical Review D*, 86(6):065007, 2012.
- [423] G. Evenbly and G. Vidal. Scaling of entanglement entropy in the (branching) multiscale entanglement renormalization ansatz. *Phys. Rev. B*, 89:235113, Jun 2014.
- [424] Ashley Milsted and Guifre Vidal. Geometric interpretation of the multi-scale entanglement renormalization ansatz, 2018.
- [425] Glen Evenbly and Guifré Vidal. Tensor network states and geometry. *Journal of Statistical Physics*, 145(4):891–918, 2011.
- [426] Justin Reyes and Miles Stoudenmire. A multi-scale tensor network architecture for classification and regression. *arXiv:2001.08286*, 2020.
- [427] Nathan A. McMahon, Sukhbinder Singh, and Gavin K. Brennen. A holographic duality from lifted tensor networks. *npj Quantum Information*, 6(1):36, Apr 2020.
- [428] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.
- [429] Ki-Seok Kim, Suk Bum Chung, Chanyong Park, and Jae-Ho Han. *Phys. Rev. D*, 99:105012, 2019.
- [430] Ki-Seok Kim, Miok Park, Jaeyoon Cho, and Chanyong Park. Emergent geometric description for a topological phase transition in the kitaev superconductor model. *Phys. Rev. D*, 96:086015, Oct 2017.
- [431] Ki-Seok Kim and Chanyong Park. Emergent geometry from field theory: Wilson’s renormalization group revisited. *Phys. Rev. D*, 93:121702, Jun 2016.
- [432] Ki-Seok Kim. Emergent dual holographic description for interacting dirac fermions in the large n limit. *Phys. Rev. D*, 102:086014, Oct 2020.
-

- [433] Ki-Seok Kim. Geometric encoding of renormalization group β -functions in an emergent holographic dual description. *Phys. Rev. D*, 102:026022, Jul 2020.
- [434] Ki-Seok Kim. Emergent geometry in recursive renormalization group transformations. *Nuclear Physics B*, 959:115144, 2020.
- [435] Ki-Seok Kim. Emergent dual holographic description for interacting dirac fermions in the large n limit. *Phys. Rev. D*, 102:086014, Oct 2020.
- [436] Ki-Seok Kim and Shinsei Ryu. Entanglement transfer from quantum matter to classical geometry in an emergent holographic dual description of a scalar field theory. *Journal of High Energy Physics*, 2021(5):260, May 2021.
- [437] Kenneth G. Wilson. The renormalization group: Critical phenomena and the kondo problem. *Rev. Mod. Phys.*, 47:773–840, Oct 1975.
- [438] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions: I. $t = 0$ phase diagram of the $1/2$ -filled hubbard model. *New Journal of Physics*, 22(6):063007, jun 2020.
- [439] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions-II: quantum criticality of the doped mott insulator. *New Journal of Physics*, 22(6):063008, jun 2020.
- [440] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.